INCH-POUND

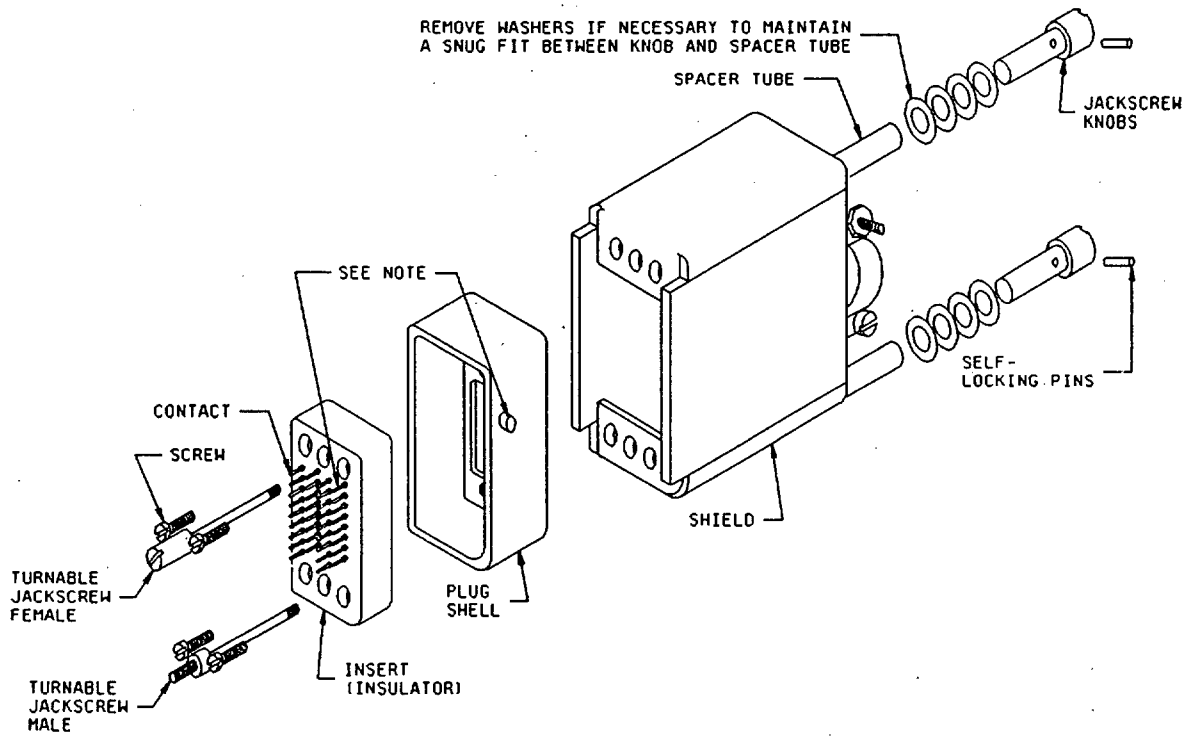
MIL-C-28748/5D
25 March 1993
SUPERSEDING
MIL-C-28748/5C
30 November 1981

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL RECTANGULAR, RACK AND PANEL,
SOLDER TYPE PIN CONTACTS, SIZE 20

This specification is approved for use by all Departments and Agencies of the department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-28748.



Note: The polarization feature (pin) on the plug shell shall be adjacent to the side closest to the number 1 or A contact position.

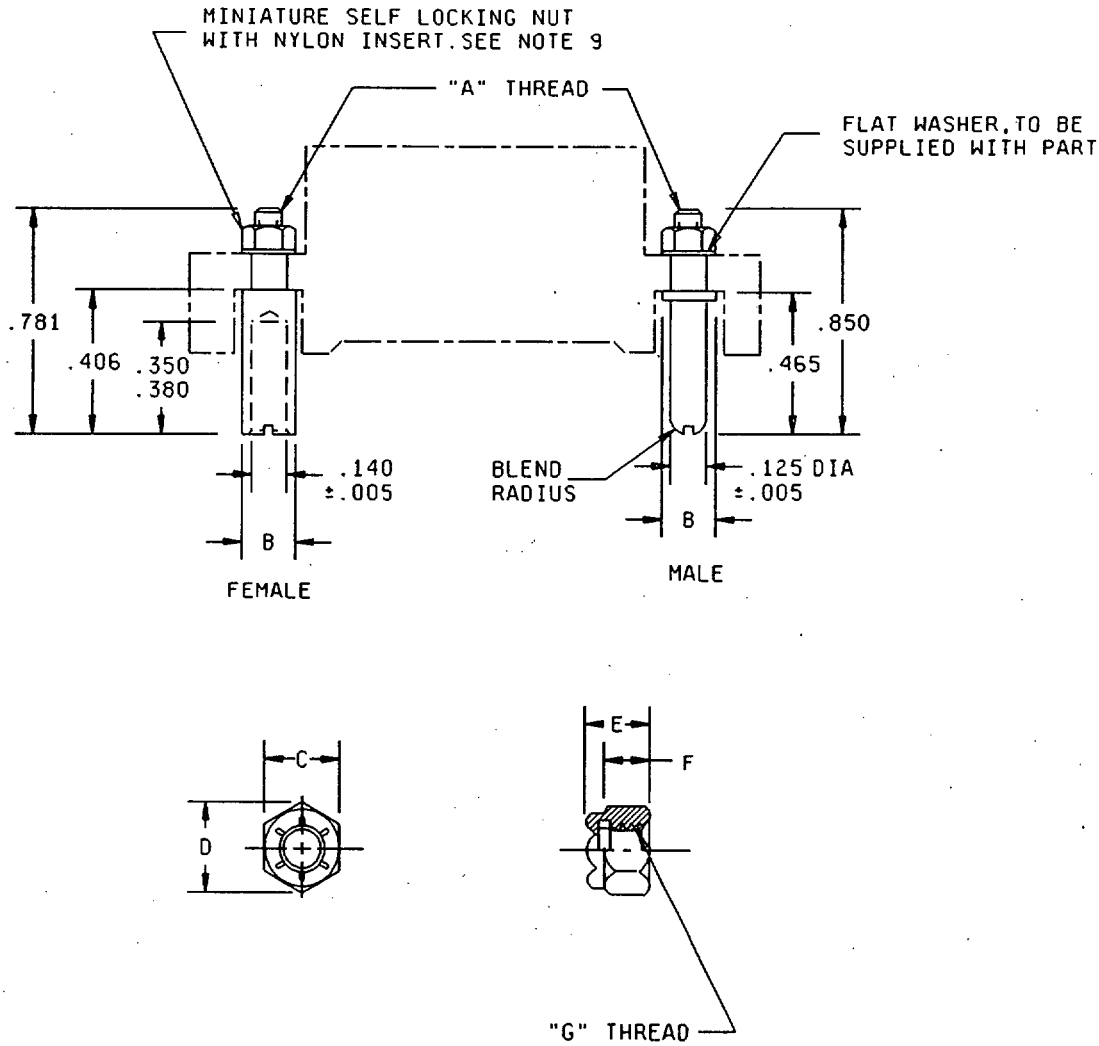
Connector

Configuration A

FIGURE 1. Connector with shields, jackscrews, and guidepins.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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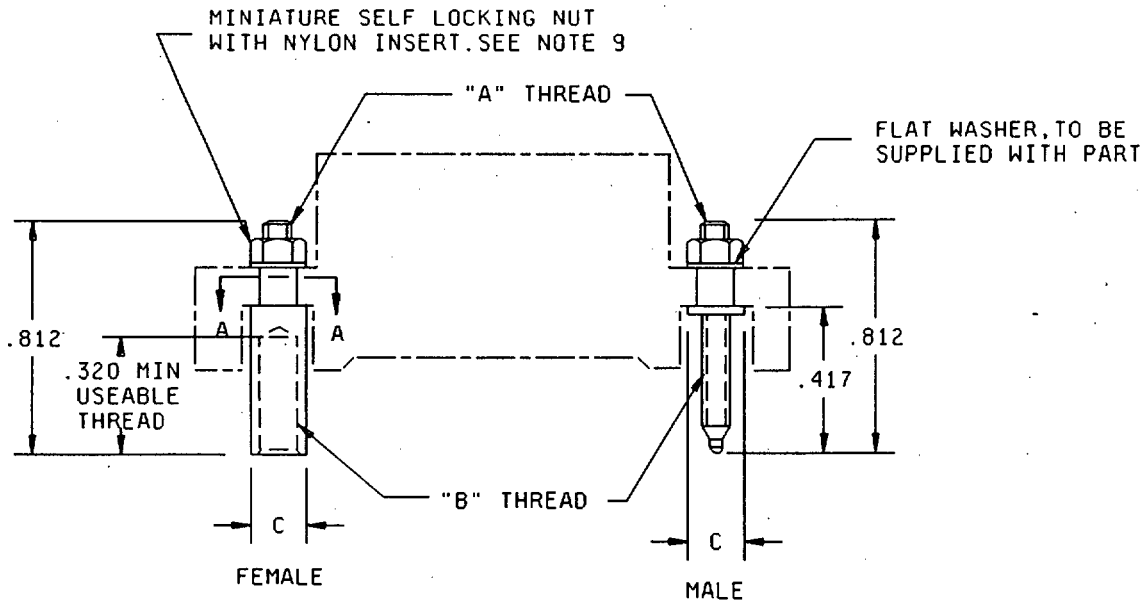
Connector size (no. of contacts)	A thread	B dia max	C max min	D ref	E max min	F ref	G thread
7 through 75	.114-40 UNC-2A	.190 (4.82)	.190 (4.82) .181 (4.59)	.205 (5.20)	.125 (3.17) .095 (2.41)	.070 (1.77)	.112-40 UNC-3B
104	.138-32 UNC-2A	.190 (4.82)	.252 (6.40) .240 (6.90)	.275 (6.98)	.153 (3.88) .133 (3.37)	.081 (2.05)	.138-32 UNC-3B

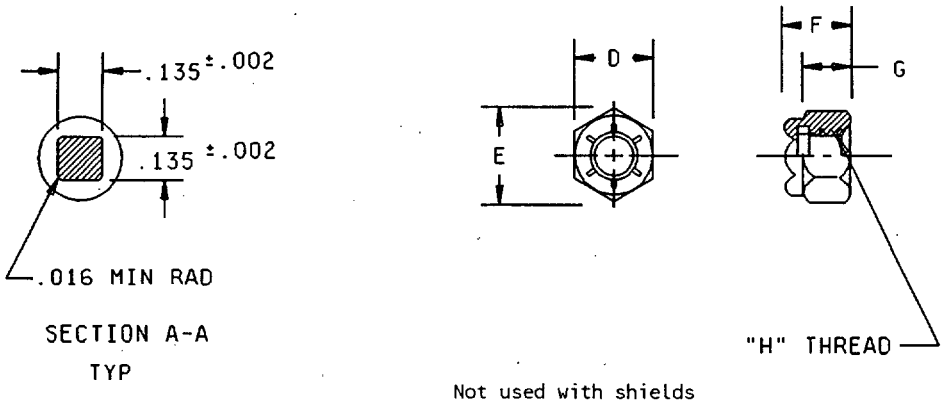
Connector with guidepins

Configuration B

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Not used with shields

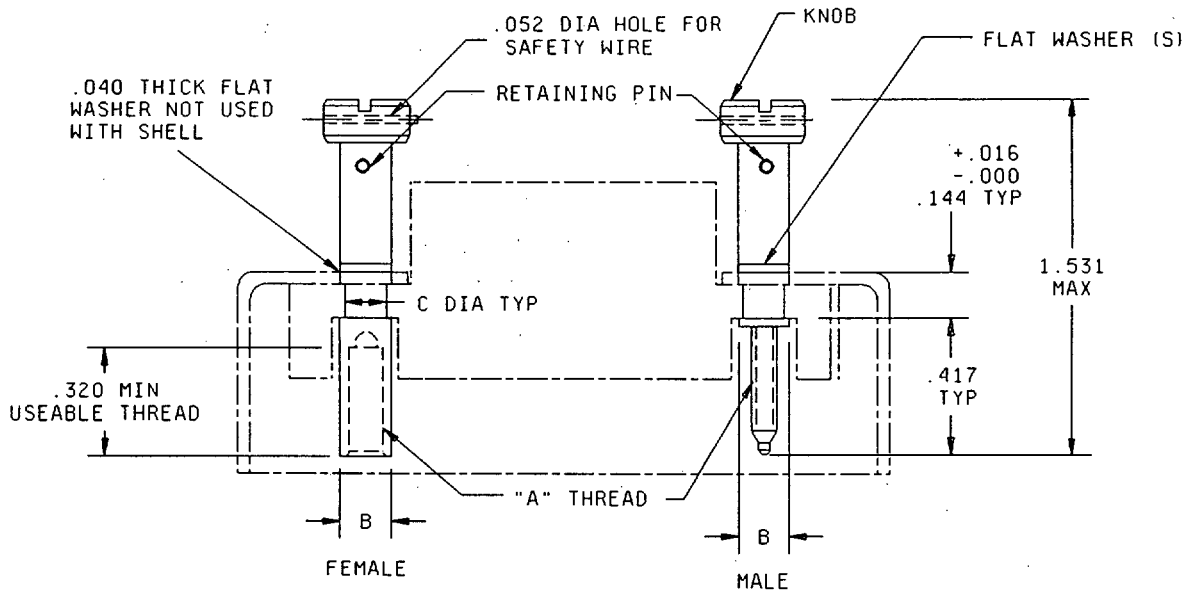
Connector size (no. of contacts)	A thread	B thread class 2	C dia max	D max min	E ref	F max min	G ref	H thread
7 through 75	.112-40 UNC-2A	.138-32 UNC	.190 (4.82)	.190 (4.82) .095 (2.41)	.205 (5.20)	.125 (3.17) .095 (2.41)	.070 (1.77)	.112-40 UNC-3B
104	.138-40 UNC-2A	.164-32 UNC	.225 (5.63)	.252 (6.40) .240 (6.09)	.275 (6.98)	.153 (1.35) .133 (3.37)	.081 (2.05)	.138-32 UNC-3B

Connector with fixed jackscrews

Configuration C

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Not used with shields.

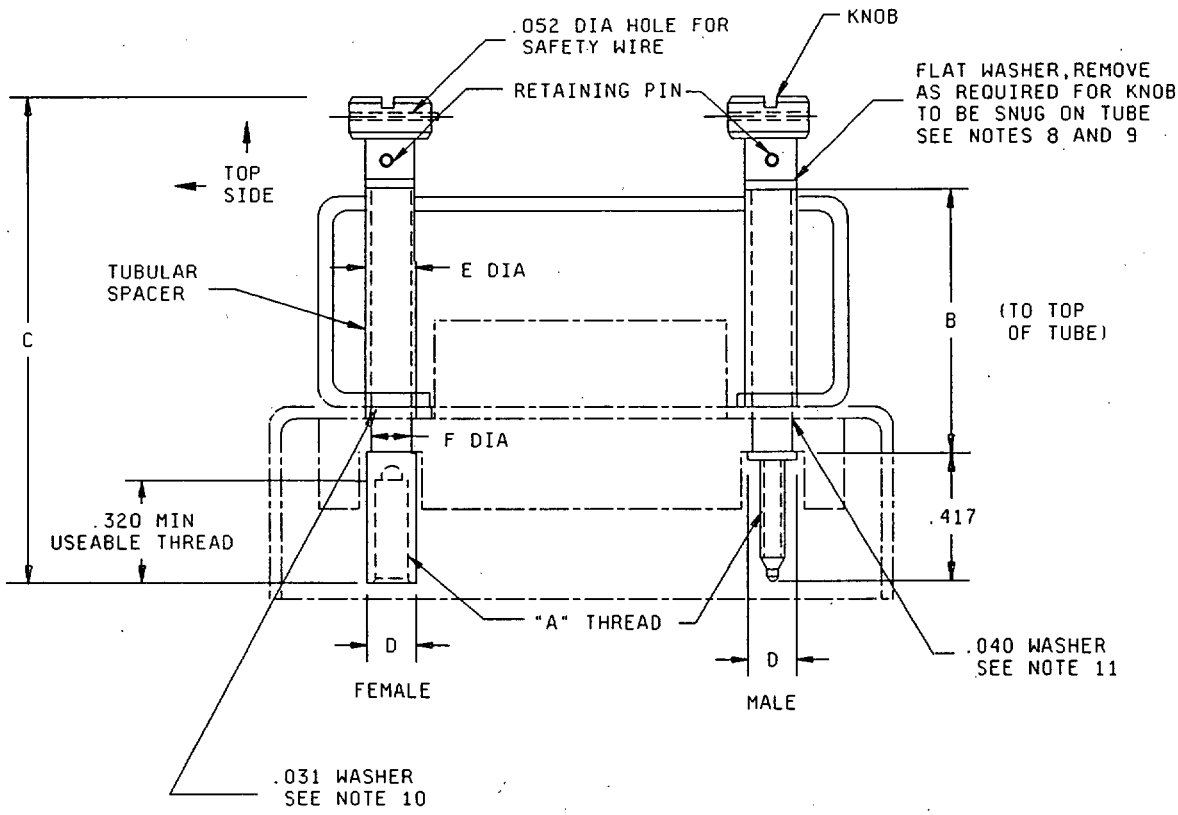
Connector size (no. of contacts)	A thread class 2	B dia. max	C dia ±.003
7 through 75	.138-32 UNC	.190 (4.82)	.115 (2.92)
104	.164-32 UNC	.225 (5.71)	.137 (4.48)

Connector with short turnable jackscrews

Configuration D

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Multipiece construction

For use with shield, configuration A; insert designator A through E, and configuration B; insert designator F through J. Plug shell; insert designator A through K, and receptacle shell; insert designator A through K.

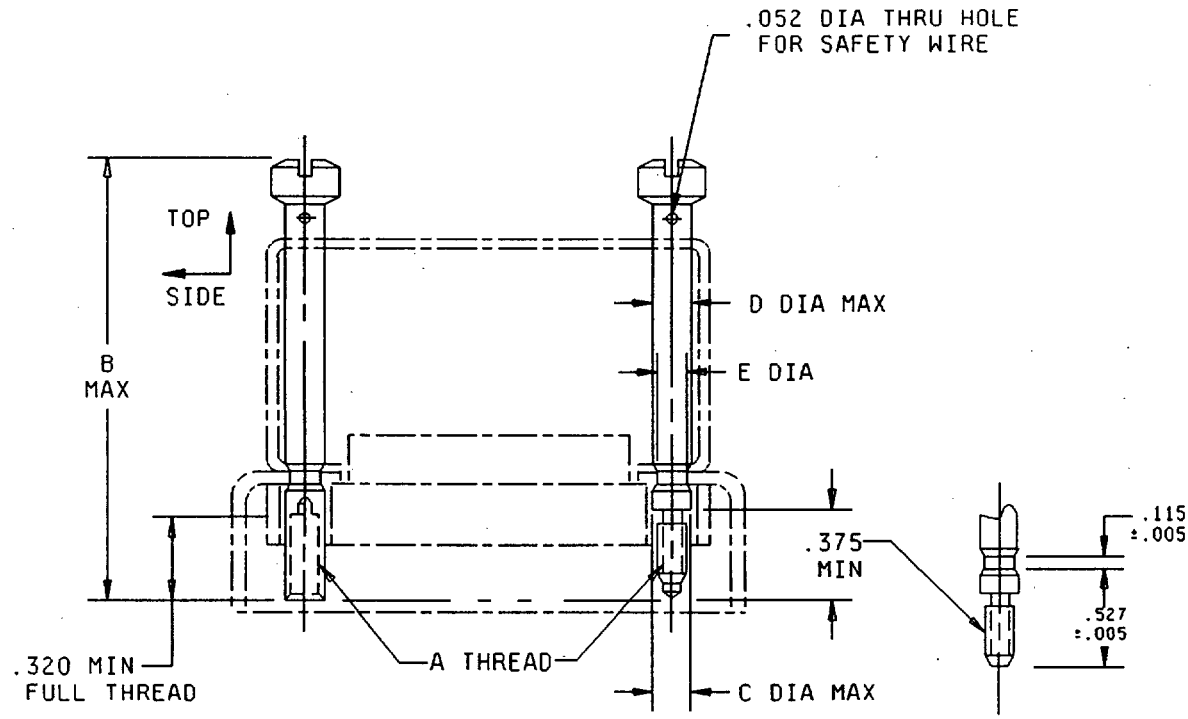
Connector size (no. of contacts)	A thread class 2	B	C max	D dia max	E dia ±.004 (0.10)	F dia ±.003 (0.07)
7 through 75	.138-32 UNC	1.500 (38.10)	2.467 (62.66)	.190 (4.82)	.188 (4.77)	.115 (2.91)
104	.164-32 UNC	2.495 (63.37)	3.438 (87.32)	.225 (5.71)	.219 (5.56)	.137 (3.47)

Connector with long turnable jackscrews and shield

Configuration E

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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One piece construction

For use with shield configurations A and B; insert designators L through Q. For use with plug and receptacle shells; insert designator L through Q.

When plug or receptacle shells; insert designators L through Q are required to be used with a shield, use with shield configuration A; insert designators A through E or shield configuration B; insert designator K.

Connector size (no. of contacts)	A thread class 2	B max	C dia max	D dia max	E dia ±.002 (0.05)
34, 42, 50, 75	.138-32 UNC	2.438 (61.93)	.192 (4.88)	.192 (4.88)	.111 (2.82)
104	.164-32 UNC	3.282 (83.36)	.225 (5.72)	.225 (5.72)	.140 (3.56)

Connector with long turnable jackscrews and shield

Configuration F

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

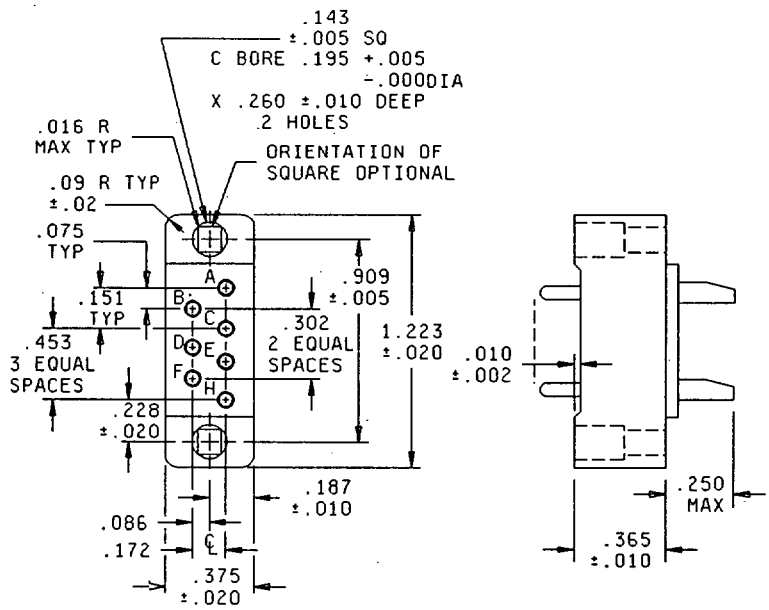
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.137	3.48	.350	8.89
.003	0.08	.138	3.51	.380	9.65
.004	0.10	.140	3.56	.406	10.31
.005	0.13	.144	3.60	.417	10.59
.016	0.41	.153	3.89	.465	11.81
.031	0.79	.181	4.60	.781	19.84
.040	1.02	.188	4.78	.812	20.62
.052	1.32	.190	4.83	.850	21.59
.070	1.78	.205	5.21	1.500	38.10
.081	2.06	.219	5.56	1.531	38.88
.095	2.41	.225	5.72	2.467	62.66
.115	2.92	.240	6.10	2.495	63.37
.125	3.18	.252	6.40	3.438	87.33
.133	3.38	.272	6.91		
.135	3.43	.320	8.13		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .016 (0.41 mm).
4. Dimensions and configuration shall be in accordance with above figure.
5. Fixed jackscrews configuration C may have .005 (0.13 mm) minimum vertical float.
6. Materials for jackscrews and guidepins: Corrosion-resistant steel in accordance with QQ-S-763, class 302, 303, 303Se, 304 or 316, ASTM A484, or ASTM A582 except for aluminum alloy knob, tubular spacer, and .040 (1.02 mm) washer.
7. Finish passivated in accordance with QQ-P-35 except for anodized knob on multipiece construction.
8. Lubricant: Solid film lubricant or "A" thread conforming to MIL-L-46010.
9. Miniature self-locking nut with nylon insert shall be in accordance with notes 6 and 7.
10. Washer .031 (0.79 mm) thick, not used with shields 20 and smaller.
11. Washer .040 (1.02 mm) thick supplied with all sizes, not used with shell.

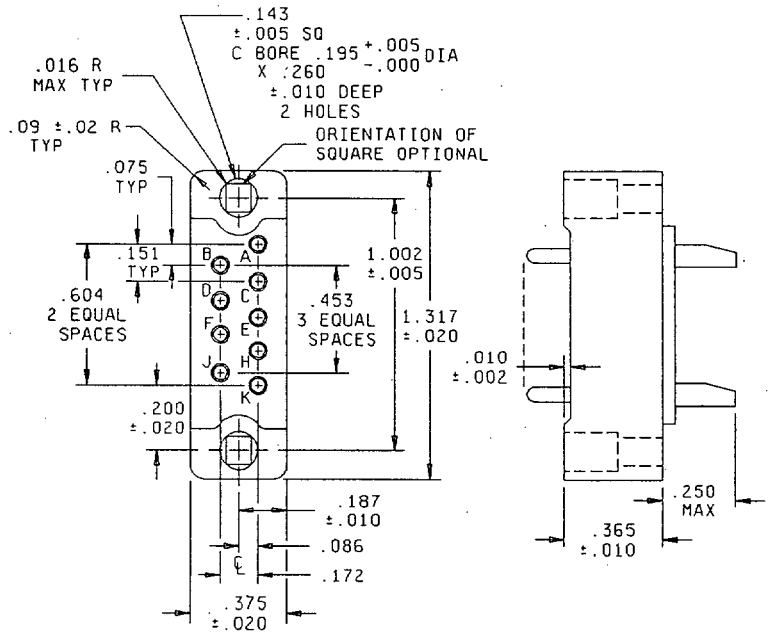
FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Insert with 7 contacts

Configuration A

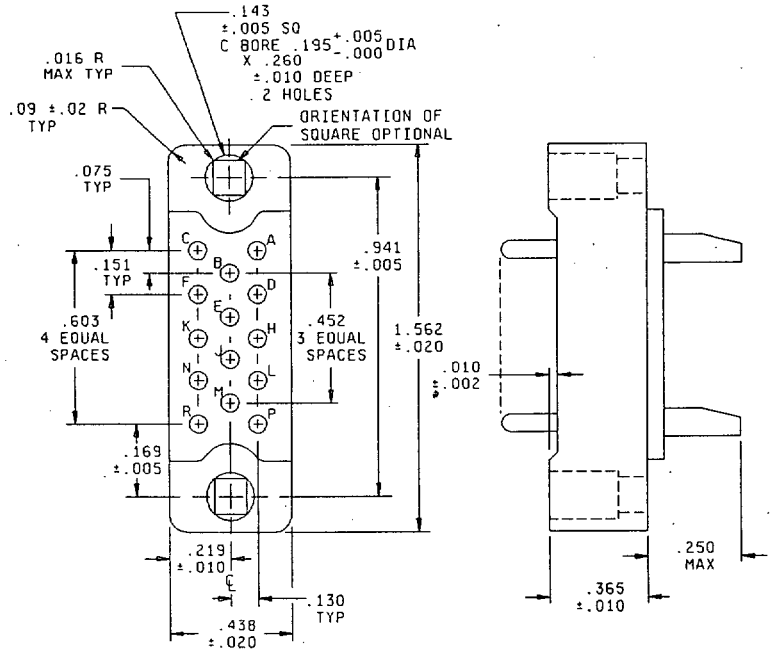


Insert with 9 contacts

Configuration B

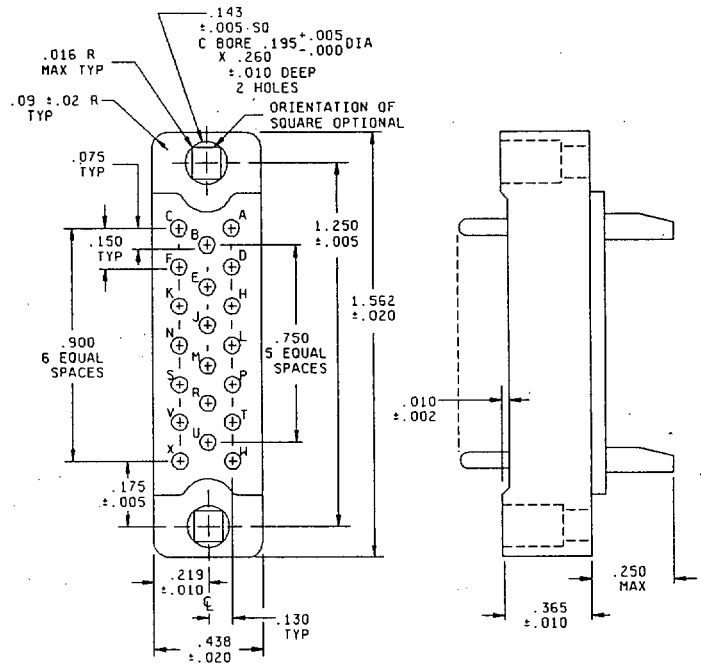
FIGURE 2. Insert arrangements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Insert with 14 contacts

Configuration C

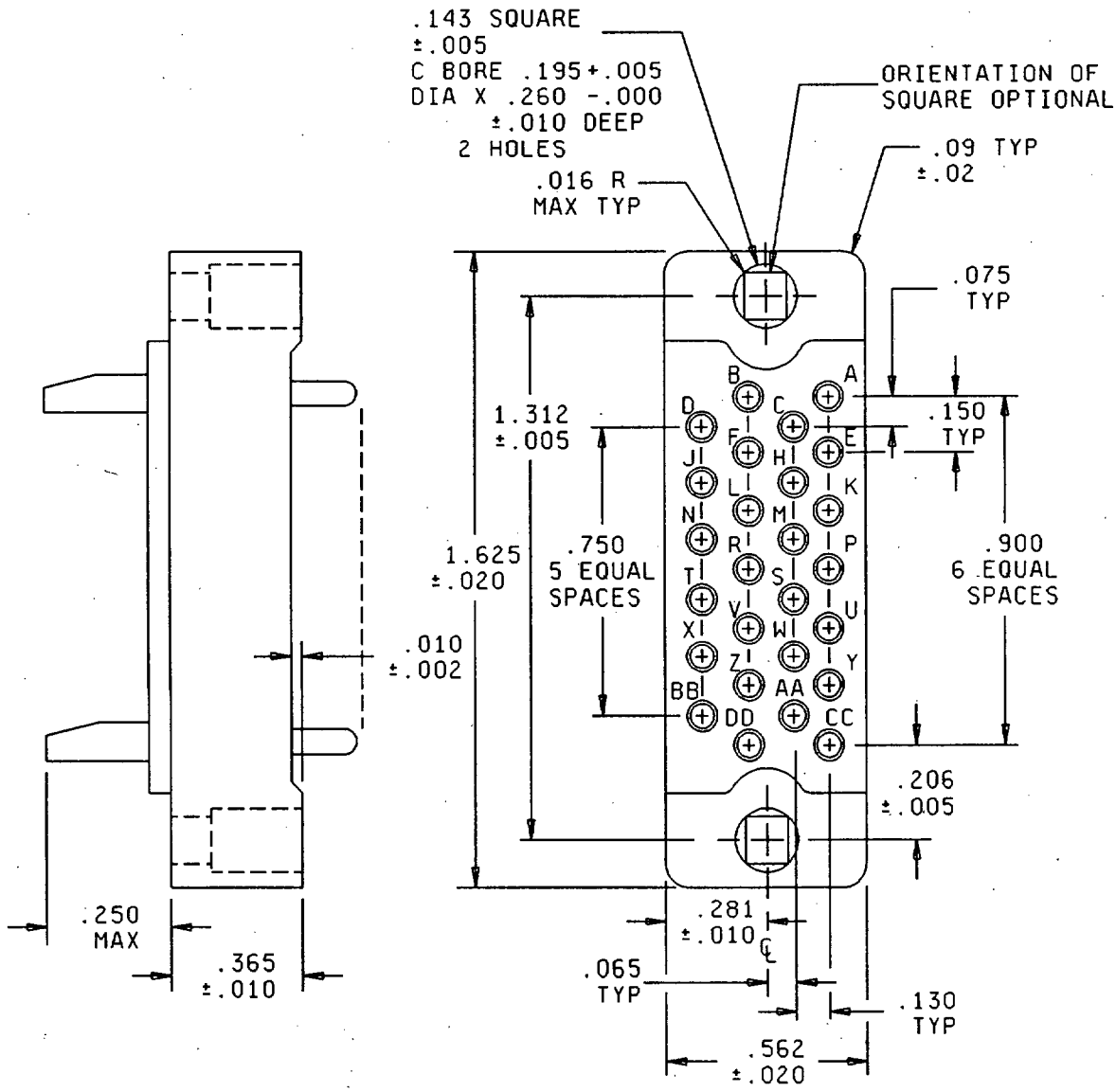


Insert with 20 contacts

Configuration D

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.

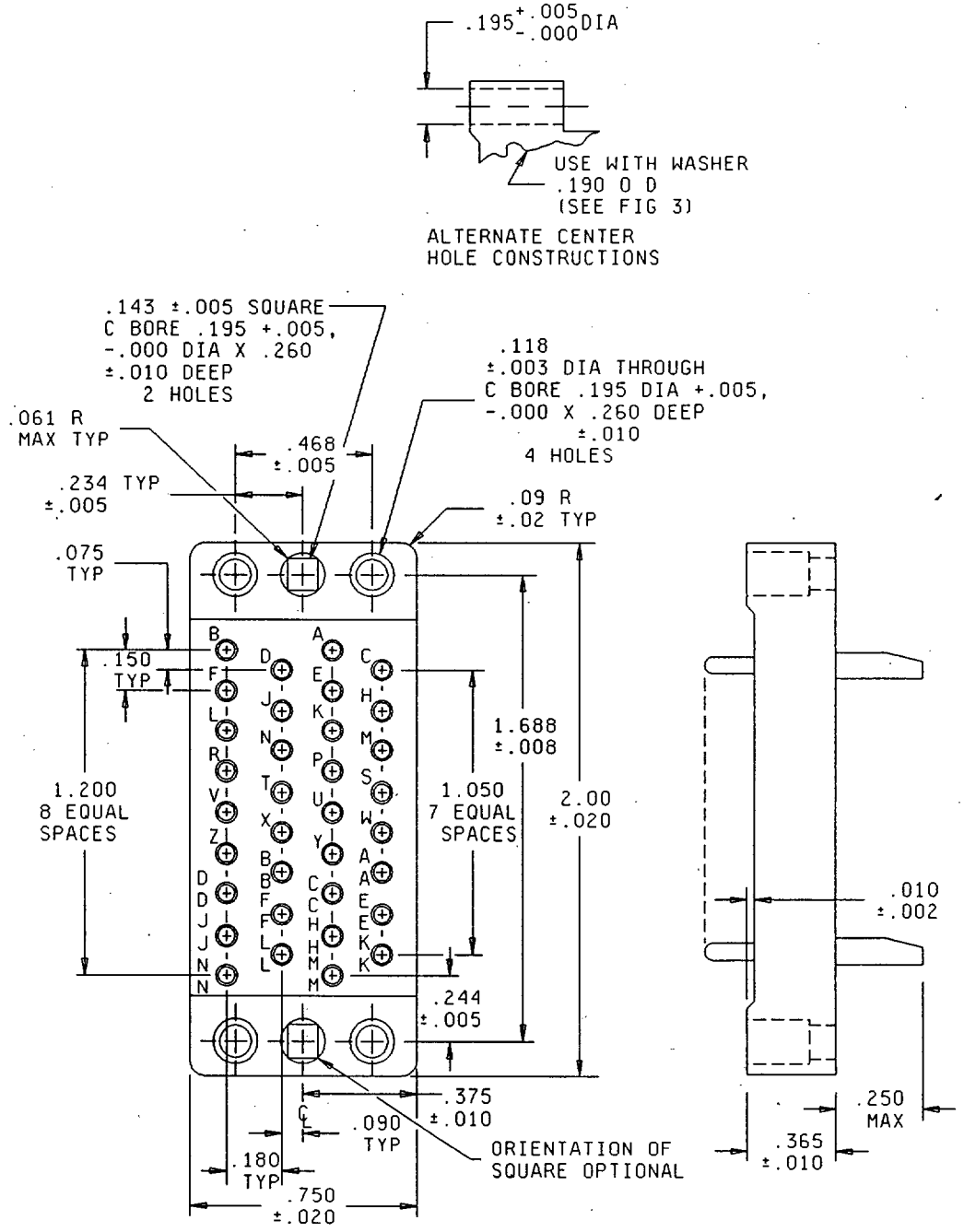


Insert with 26 contacts

Configuration E

FIGURE 2. Insert arrangements - Continued.

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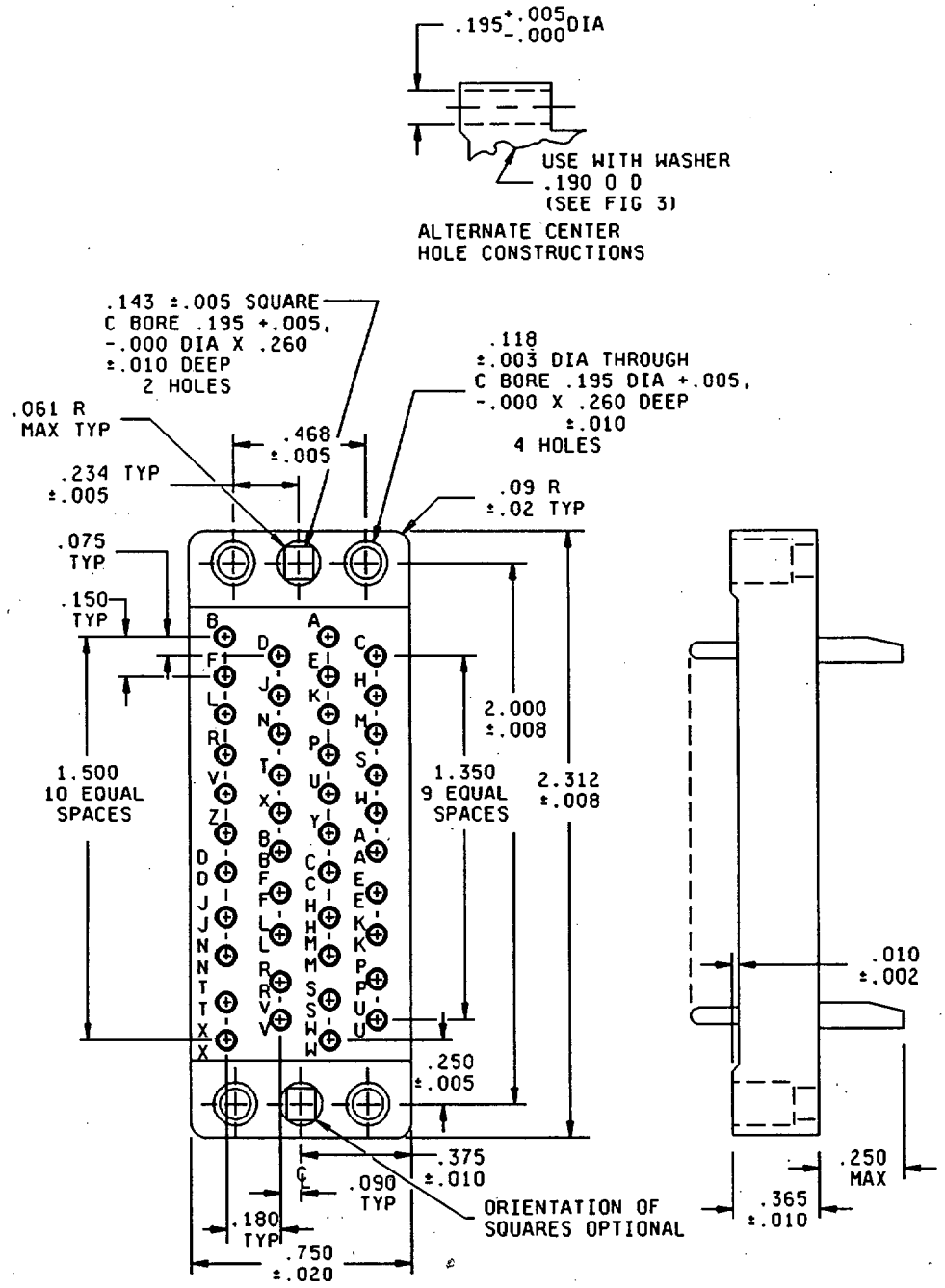
Configuration Q (X) for long jackscrews and hood only (thru hole alternate construction)

Insert with 34 contacts

Configurations F (W) and Q (X)

FIGURE 2. Insert arrangements - Continued.

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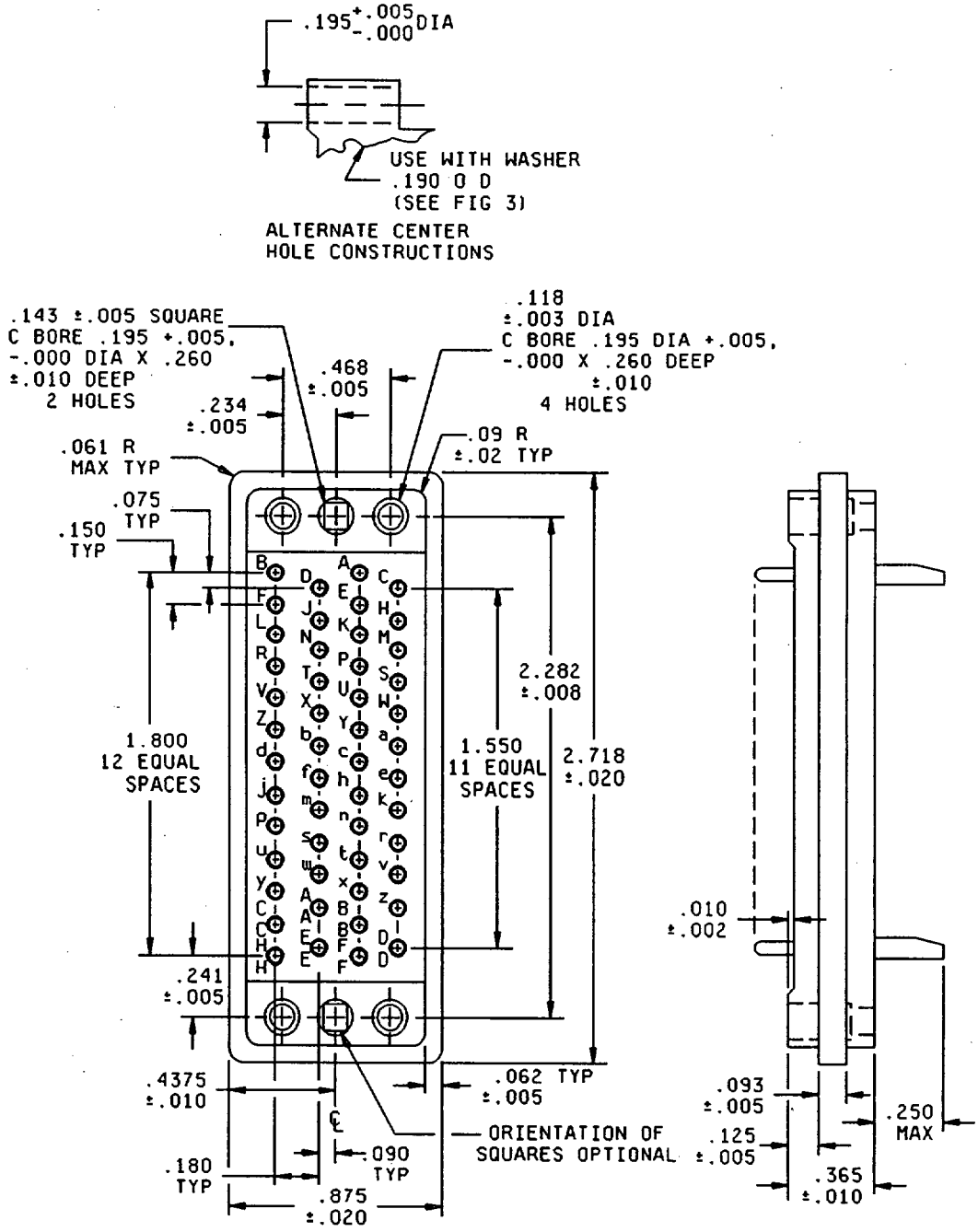
Configuration P (Z) for long jackscrews and hood only (thru hole alternate construction)

Insert with 42 contacts

Configurations G (Y) and P (Z)

FIGURE 2. Insert arrangements - Continued.

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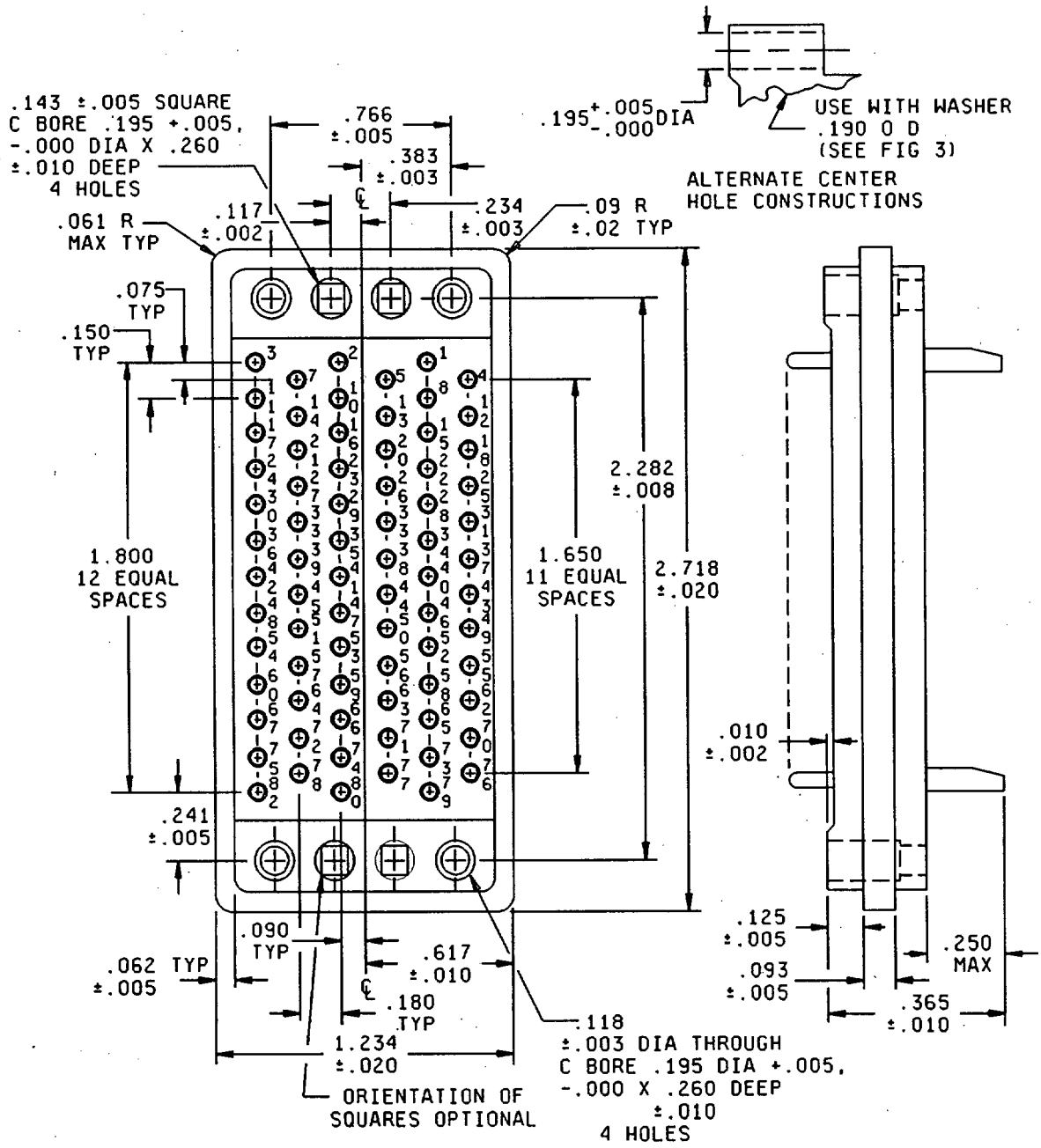
Configuration N (2) for long jackscrews and hood only (thru hole alternate construction)

Insert with 50 contacts

Configurations H (1) and N (2)

FIGURE 2. Insert arrangements - Continued.

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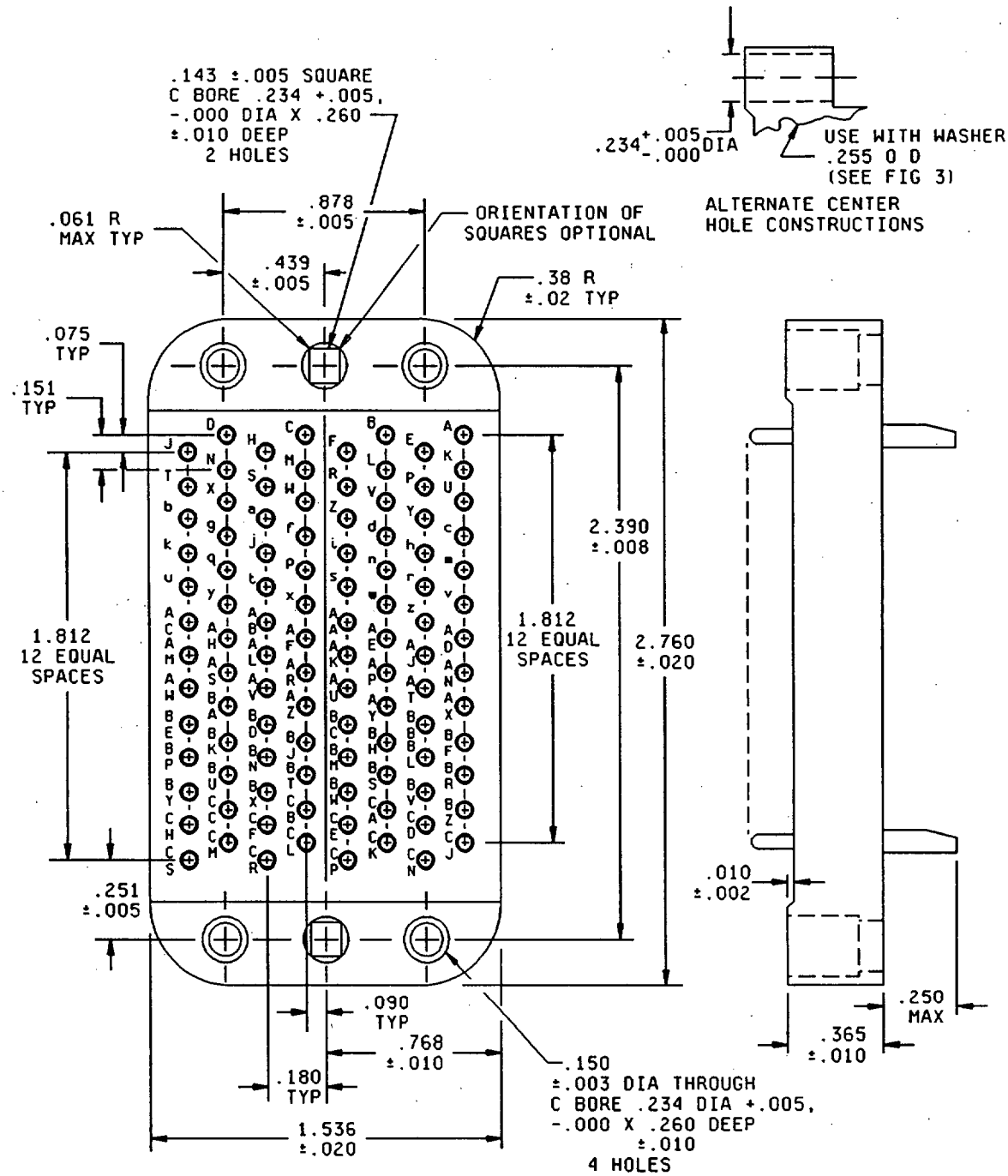
Configuration M (4) for long jackscrews and hood only (thru hole alternate construction)

Insert with 75 contacts

Configurations J (3) and M (4)

FIGURE 2. Insert arrangements - Continued.

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Configuration L (6) for long jackscrews and hood only (thru hole alternate construction)

Insert with 104 contacts

Configurations K (5) and L (6)

FIGURE 2. Insert arrangements - Continued.

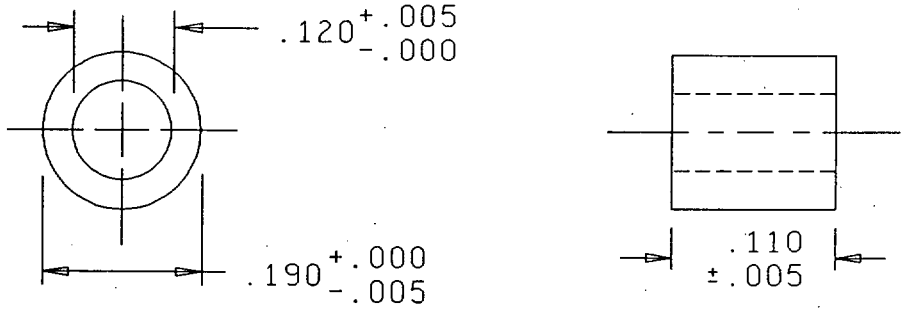
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.153	3.89	.302	7.67	.909	23.09
.003	0.08	.160	4.06	.350	8.89	1.002	25.45
.005	0.13	.172	4.37	.365	9.27	1.050	26.67
.008	0.20	.175	4.45	.375	9.53	1.200	30.48
.010	0.25	.180	4.57	.380	9.65	1.223	31.06
.016	0.41	.181	4.60	.406	10.31	1.234	31.34
.020	0.51	.187	4.75	.4375	11.113	1.250	31.75
.062	1.57	.190	4.83	.438	11.13	1.317	33.45
.065	1.65	.195	4.95	.439	11.15	1.350	34.29
.070	1.78	.200	5.08	.452	11.48	1.500	38.10
.075	1.91	.205	5.21	.453	11.51	1.536	39.01
.081	2.06	.206	5.23	.465	11.81	1.562	39.67
.086	2.18	.219	5.56	.468	11.89	1.625	41.28
.090	2.29	.228	5.79	.562	14.27	1.650	41.91
.093	2.36	.234	5.94	.603	15.32	1.688	42.67
.095	2.41	.240	6.10	.617	15.63	1.800	45.72
.110	2.79	.241	6.12	.750	19.05	2.000	50.80
.118	3.00	.244	6.20	.766	19.46	2.282	57.96
.125	3.18	.250	6.35	.768	19.51	2.312	57.72
.130	3.30	.251	6.38	.781	19.84	2.390	60.71
.133	3.38	.252	6.40	.850	21.59	2.718	69.04
.143	3.63	.260	6.60	.875	22.23	2.760	70.10
.150	3.81	.272	6.91	.878	22.30		
.151	3.84	.281	7.14	.900	22.86		

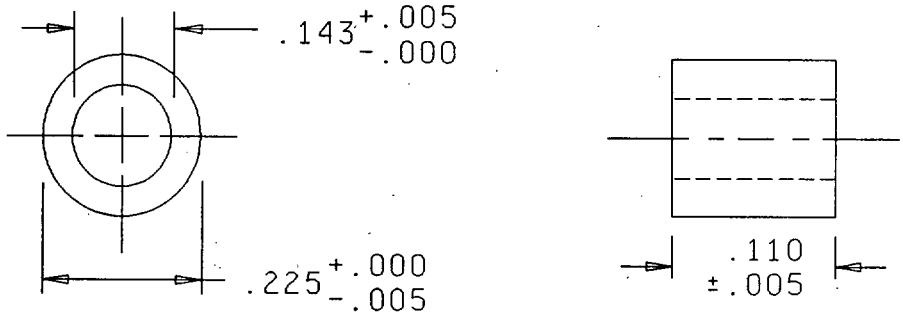
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm).
4. Tolerance between any two adjacent contact centers shall be ±.004 (0.10 mm). Tolerance between any two contact centers, other than adjacent centers, shall be ±.006 (0.15 mm).
5. Dimensions and configuration shall be in accordance with above figure.
6. Contact identification may be located on either side of the contact hole but shall appear on the front and rear of each insert.

FIGURE 2. Insert arrangements - Continued.

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For use with inserts with 34, 42, 50, and 75 contacts



For use with inserts with 104 contacts.

Inches	mm
.005	0.13
.110	2.79
.120	3.05
.143	3.63
.190	4.83
.225	5.72

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .015 (0.38 mm).
3. Material: Corrosion resistant steel, in accordance with QQ-S-763 or QQ-S-764, class 32, 303, 303Se, 304 or 316.
4. Finish: Passivated in accordance with QQ-P-35.

FIGURE 3. Washer.

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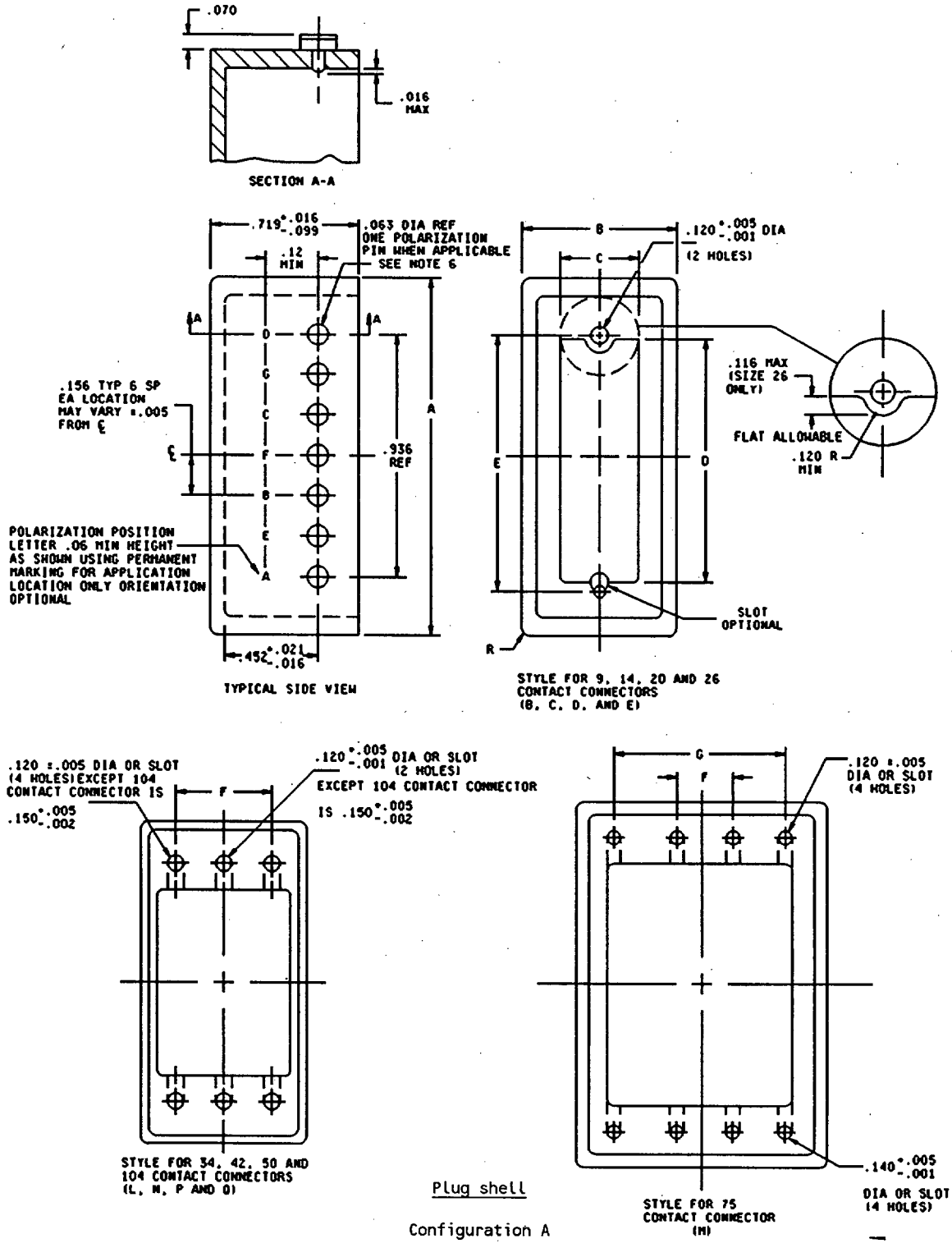
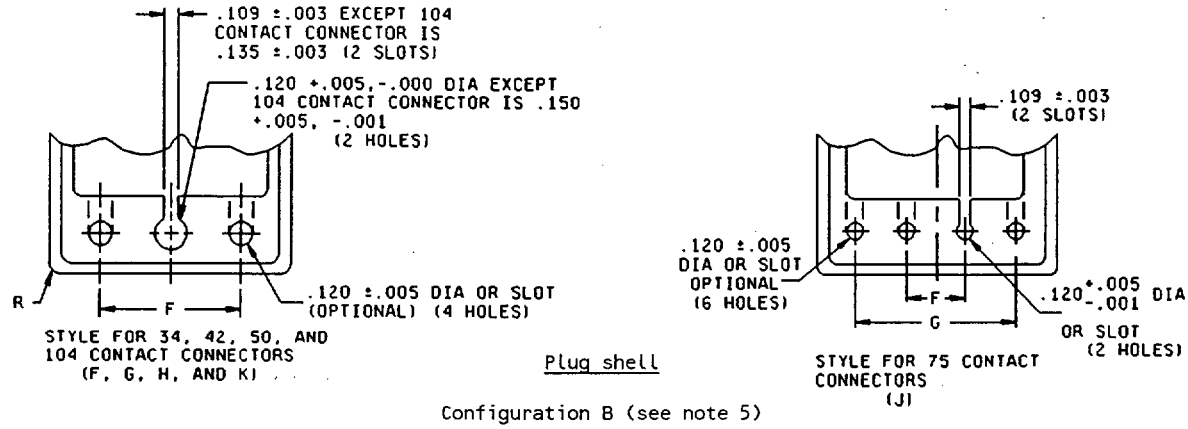


FIGURE 4. Shell configurations.

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Plug shell

Configuration B (see note 5)

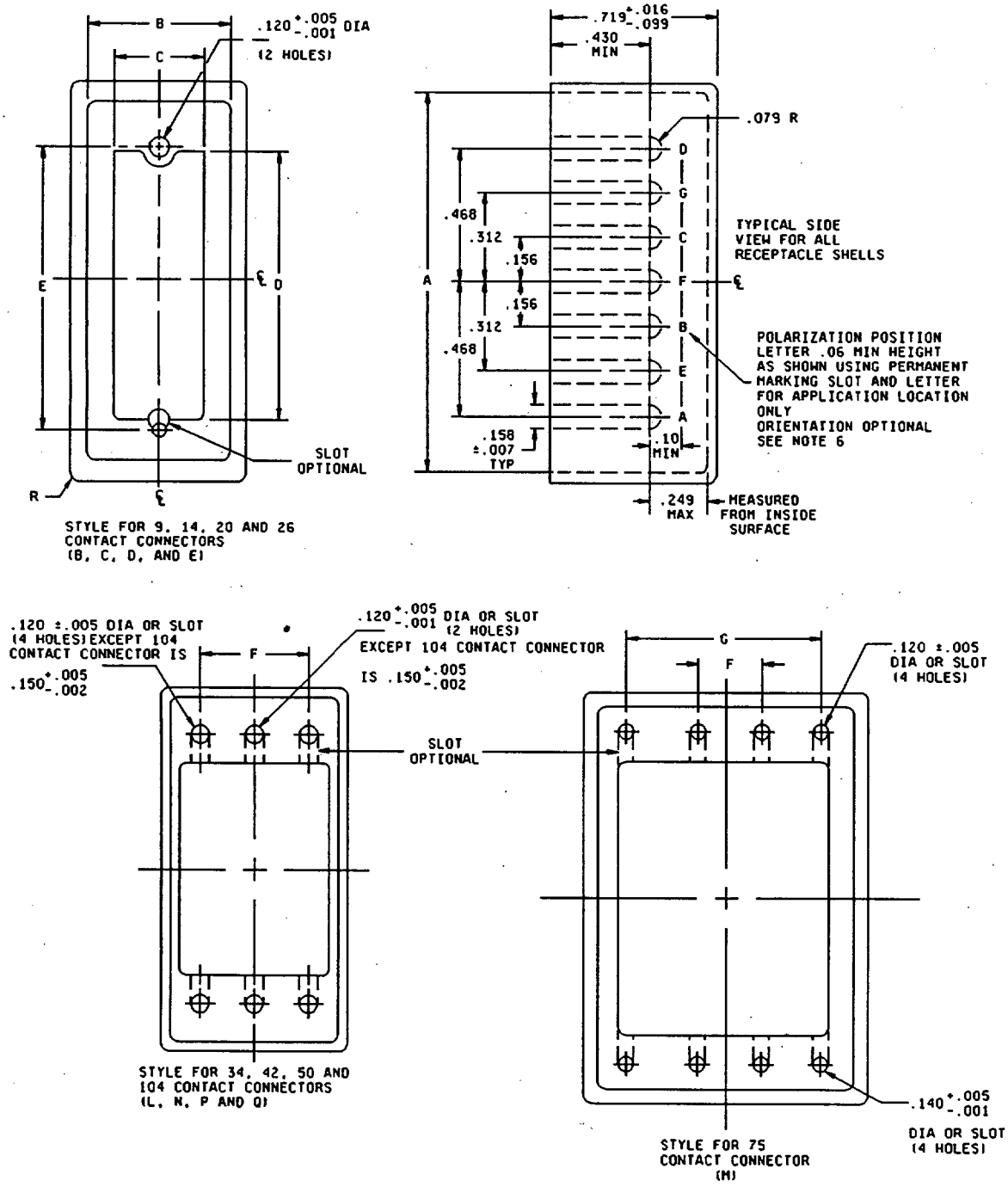
Insert designator (see note 7)	Connector size (number of contacts)	A max	B max	C min	D min	E ±.005 (0.13)	F ±.005 (0.13)	G ±.005 (0.13)	R
B (S)	9	1.440 (36.57)	.500 (12.70)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	-----	-----	.141 (3.58)
C (T)	14	1.380 (35.05)	.567 (14.40)	.440 (11.17)	.800 (20.32)	.938 (23.82)	-----	-----	.141 (3.58)
D (U)	20	1.700 (43.18)	.567 (14.40)	.440 (11.17)	1.097 (27.86)	1.250 (31.75)	-----	-----	.141 (3.58)
E (V)	26	1.755 (44.57)	.688 (17.47)	.565 (14.35)	1.182 (30.02)	1.312 (33.32)	-----	-----	.141 (3.58)
F (W) or Q (X)	34	2.130 (54.10)	.880 (22.35)	.745 (18.92)	1.400 (35.56)	1.688 (42.87)	.468 (11.88)	-----	.141 (3.58)
G (Y) or P (Z)	42	2.442 (62.02)	.880 (22.35)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.88)	-----	.141 (3.58)
H (1) or N (2)	50	2.846 (72.28)	1.005 (25.52)	.745 (18.92)	2.026 (51.46)	2.281 (57.93)	.468 (11.88)	-----	.141 (3.58)
J (3) or M (4)	75	2.850 (72.39)	1.365 (34.67)	1.104 (28.04)	2.026 (51.46)	2.281 (57.93)	.234 (5.94)		.187 (4.74)
K (5) or L (6)	104	2.890 (73.40)	1.670 (42.18)	1.476 (37.49)	2.073 (52.50)	2.375 (60.32)	.875 (22.22)	-----	.422 (10.71)

Plug shell

Configurations A and B

FIGURE 4. Shell configurations - Continued.

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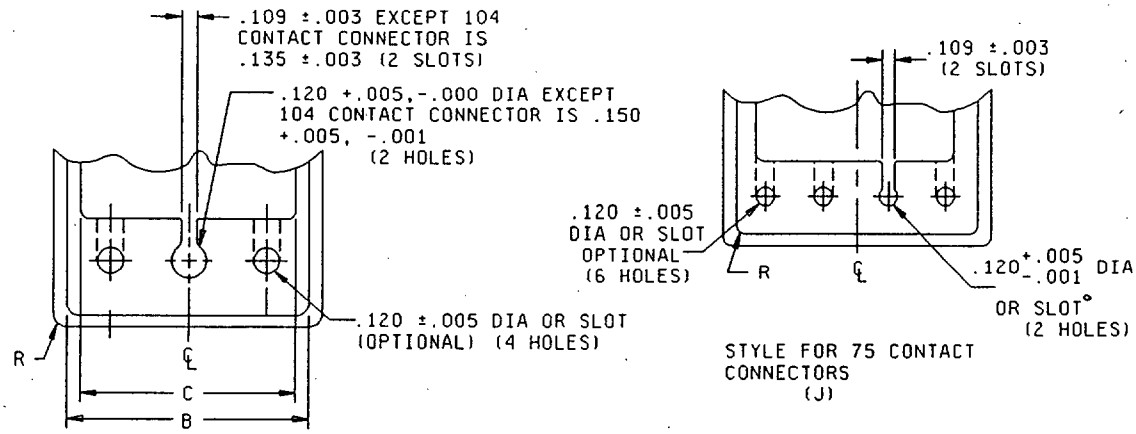


Receptacle shell

Configuration A

FIGURE 4. Shell configurations - Continued.

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STYLE FOR 34, 42, 50, AND 104 CONTACT CONNECTORS (F, G, H, AND K)

Receptacle shell

Configuration B (see note 5)

Insert designator (see note 7)	Connector size (number of Contacts)	A min	B min	C min	D min	E ±.005 (0.13)	F ±.005 (0.13)	G ±.005 (0.13)	R
B	9	1.448 (36.77)	.512 (13.00)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	-----	-----	.141 (3.58)
C	14	1.388 (35.25)	.575 (14.60)	.440 (11.17)	.800 (20.32)	.938 (23.82)	-----	-----	.141 (3.58)
D	20	1.702 (43.23)	.575 (14.60)	.440 (11.17)	1.097 (27.86)	1.250 (31.75)	-----	-----	.141 (3.58)
E	26	1.763 (44.78)	.696 (17.67)	.565 (14.35)	1.182 (30.02)	1.312 (33.32)	-----	-----	.141 (3.58)
F (W) or Q (X)	34	2.138 (54.30)	.888 (22.55)	.745 (18.92)	1.400 (35.56)	1.688 (42.87)	.468 (11.88)	-----	.141 (3.58)
G (Y) or P (Z)	42	2.450 (62.23)	.888 (22.55)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.88)	-----	.141 (3.58)
H (1) or N (2)	50	2.856 (72.54)	1.015 (25.78)	.745 (18.92)	2.026 (51.46)	2.281 (57.93)	.468 (11.88)	-----	.188 1/ (4.77)
J (3) or M (4)	75	2.855 (72.51)	1.370 (34.79)	1.104 (28.04)	2.026 (51.46)	2.281 (57.93	.234 (5.94)	.766 (19.45)	.203 (5.15)
K (5) or L (6)	104	2.900 (73.66)	1.680 (42.67)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)	-----	.422 (10.71)

1/ Tolerance +.020 inch (0.50 mm), -.016 inch (0.40 mm).

Receptacle shell

Configurations A and B

FIGURE 4. Shell configurations - Continued.

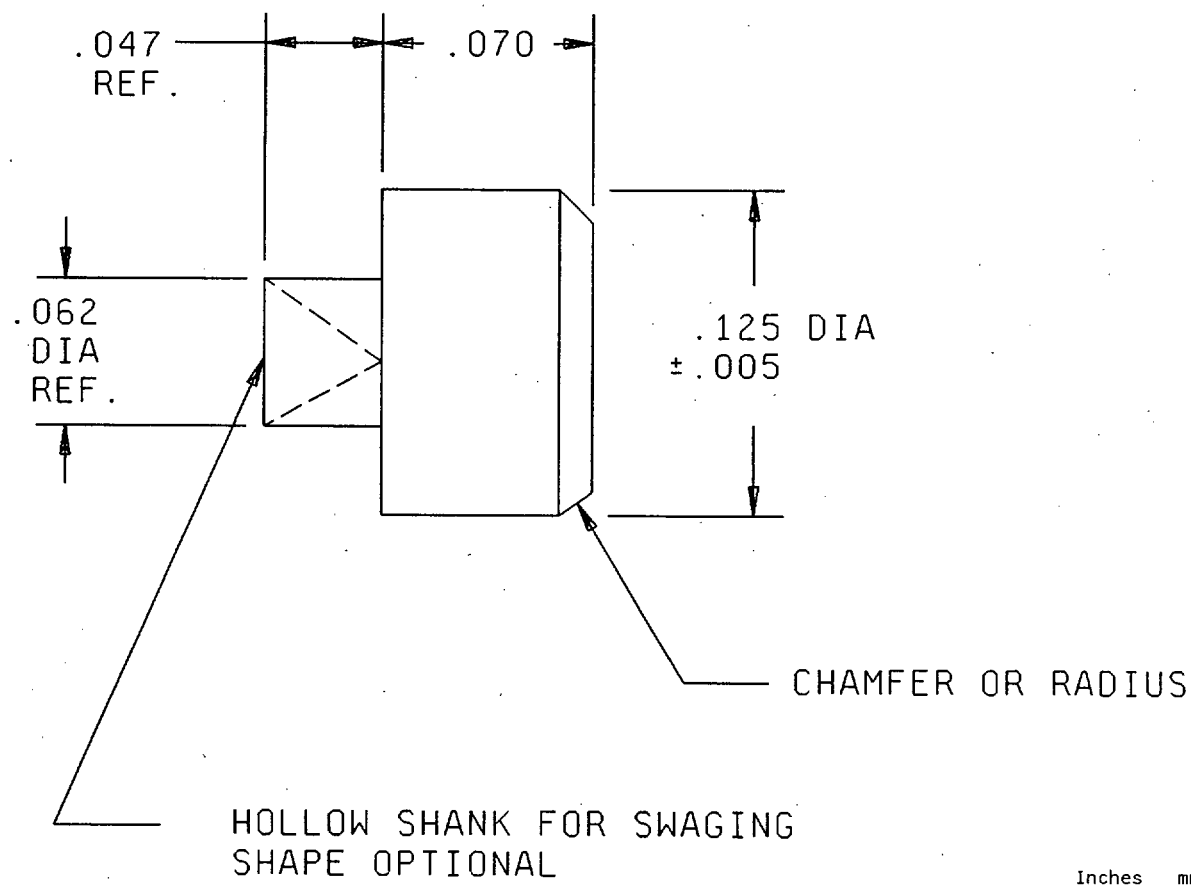
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.187	4.75	.766	19.46	1.380	35.05	2.420	61.47
.002	0.05	.188	4.78	.800	20.32	1.400	35.56	2.442	62.03
.003	0.08	.203	5.16	.864	21.94	1.440	36.58	2.450	62.23
.005	0.13	.234	5.94	.875	22.22	1.448	3 78	2.846	72.29
.007	0.18	.249	6.32	.880	22.35	1.476	37.49	2.850	72.39
.016	0.41	.312	7.92	.888	22.56	1.670	42.42	2.855	72.52
.020	0.51	.375	9.52	.936	23.77	1.680	42.67	2.856	72.54
.021	0.53	.422	10.72	.938	23.82	1.688	42.88	2.890	73.41
.060	1.52	.430	10.92	1.000	25.40	1.696	43.08	2.900	73.66
.099	2.51	.440	11.18	1.005	25.53	1.700	43.18		
.100	2.54	.452	11.48	1.015	25.78	1.702	43.23		
.104	2.64	.468	11.89	1.097	27.86	1.755	44.58		
.109	2.77	.500	12.70	1.104	28.04	1.969	50.01		
.116	2.95	.512	13.00	1.115	28.32	2.000	50.80		
.120	3.05	.565	14.35	1.182	30.02	2.026	51.46		
.135	3.43	.567	14.40	1.250	31.75	2.073	52.65		
.140	3.56	.575	14.60	1.254	31.85	2.130	54.10		
.141	3.58	.688	17.48	1.264	32.10	2.138	54.30		
.150	3.81	.696	17.68	1.312	33.32	2.281	57.94		
.156	3.96	.719	18.26	1.365	34.67	2.375	60.32		
.158	4.01	.745	18.92	1.370	34.80	2.410	61.21		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
3. Material: Aluminum alloy in accordance with QQ-A-250/8, temper O, or QQ-A-250/1, temper H12, .040 inch (1.02 mm).
4. Finish: Anodize in accordance with MIL-A-8625, type I or type II.
5. Not used with one piece jackscrews shields of figure 2, configuration B.
6. The part number includes the polarization letter from figure. Use the letter "R" if polarization is not required.
7. Insert designators in parentheses are for piece part ordering data.

FIGURE 4. Shell configurations - Continued.

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Inches	mm
.005	0.13
.047	1.19
.062	1.57
.070	1.78
.125	3.18

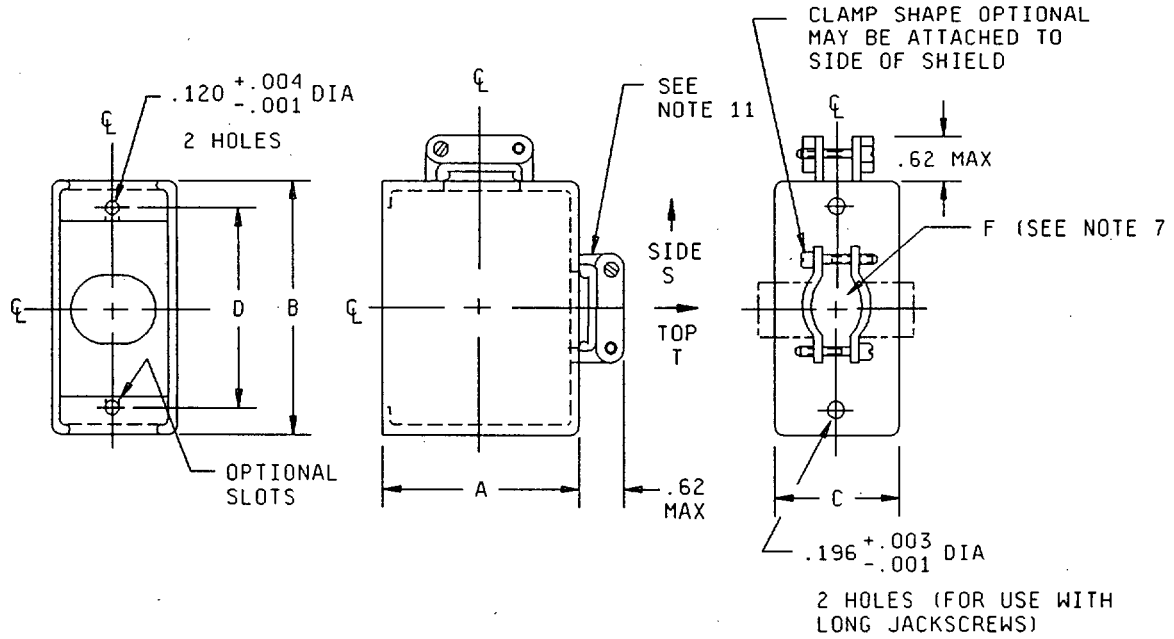
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
3. Metric equivalents are given for general information only.
4. Material: Corrosion resistant steel in accordance with QQ-S-763, class 302 303, 303Se, 304 or 316, or ASTM A484, or ASTM A532.

Pin, straight headed

Configuration

FIGURE 5. Pin configuration.

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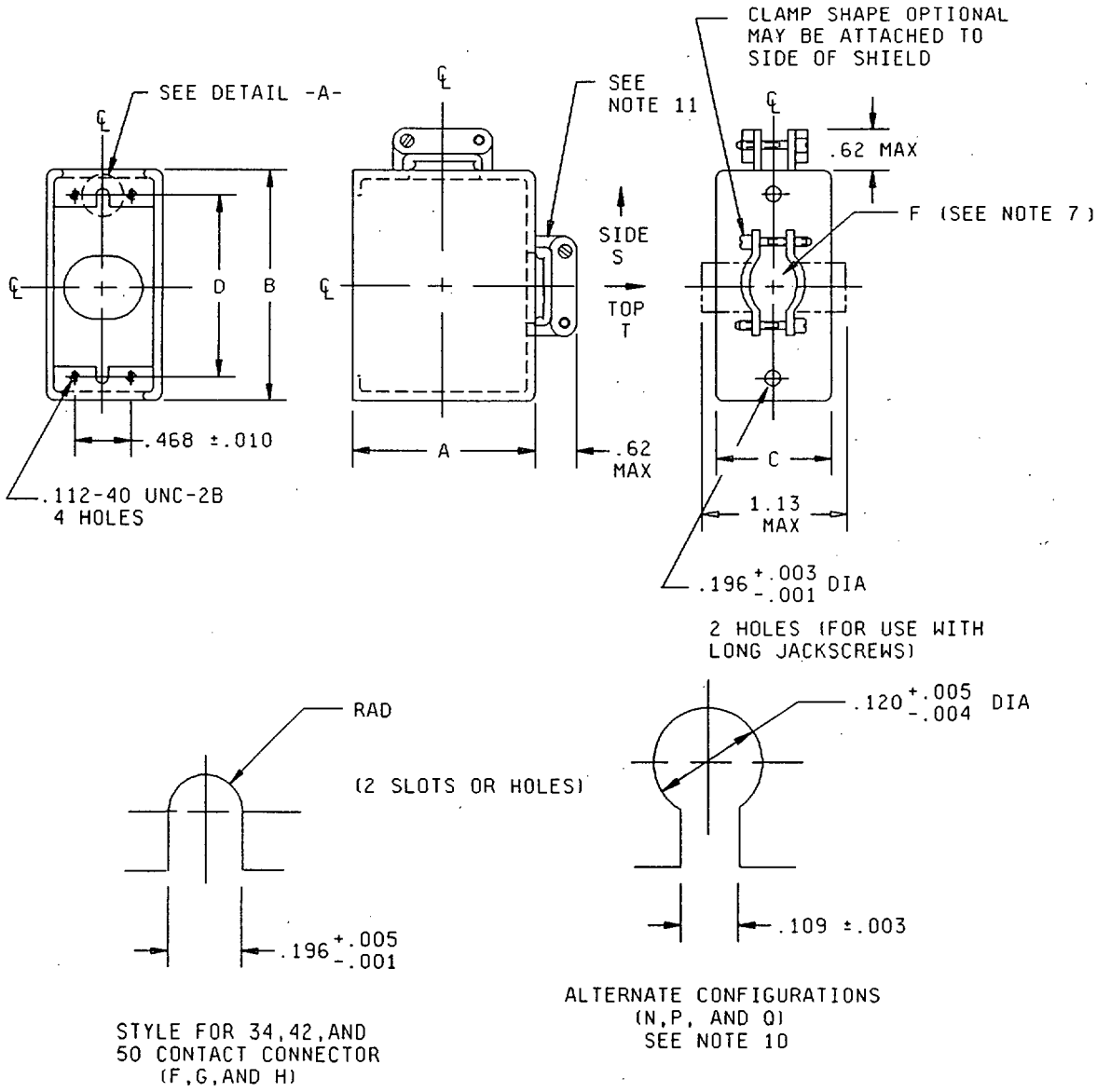
For use with insert configurations A (R), B (S), C (T), D (U), and E (V)

Shield

Configuration A

FIGURE 6. Shield configurations.

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Check the source to verify that this is the current version before use.



DETAIL -A-

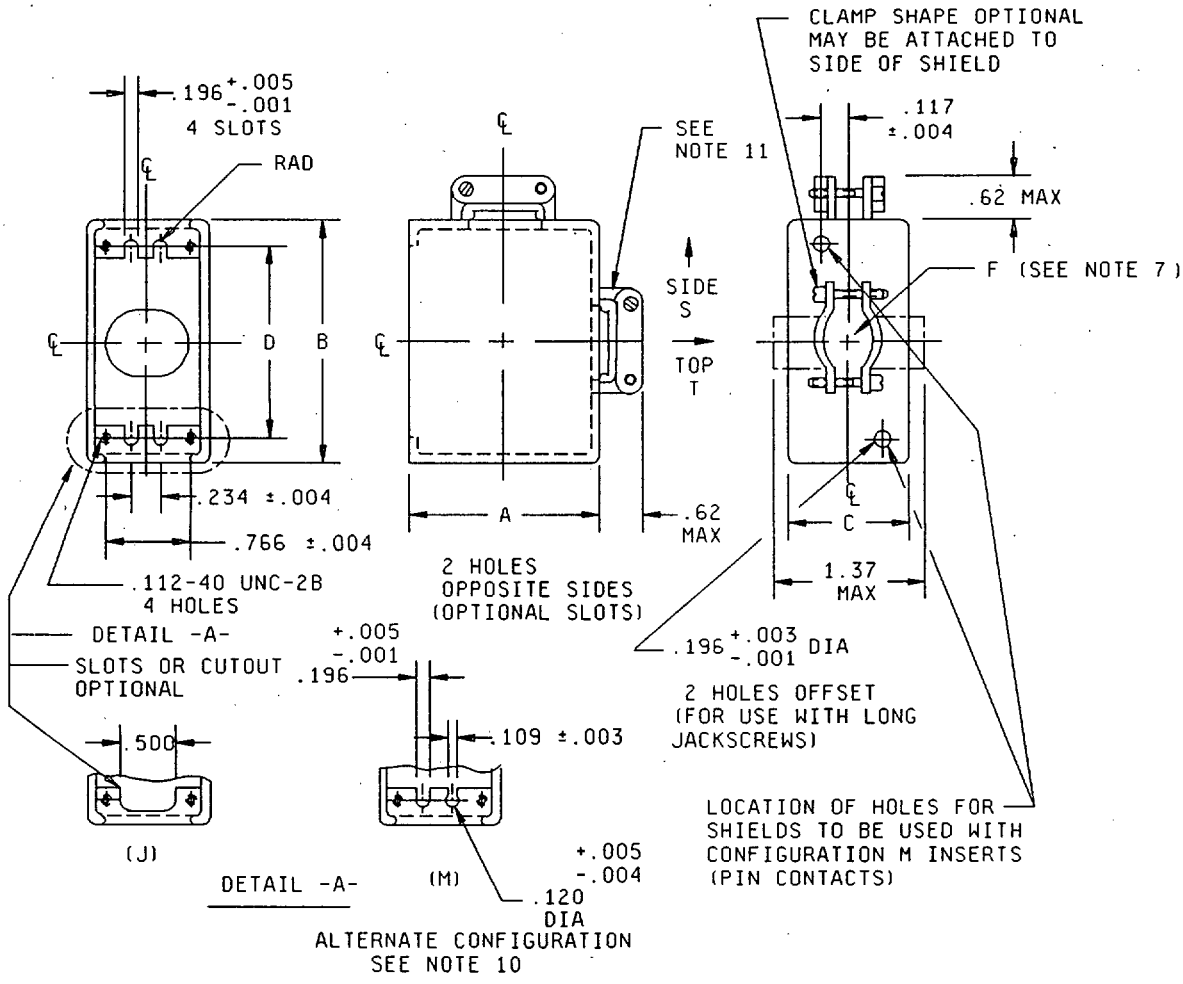
For use with insert configurations F (W), G (Y), H (1), N (2), P (Z), and Q (X)

Shield

Configuration A

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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For use with insert configurations J (3) and M (4)

Shield

Configuration A

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

Insert designator 1/ 2/ 3/	Configuration and cable opening	Connector size	A max min	B ±.016 (.41)	C ±.016 (.41)	D ±.010 (.25)	F min cable opening
A (R)	1 or 2	7	1.35 (34.3) 1.18 (30.0)	1.220 (30.99)	.437 (11.09)	.906 (23.01)	.18 (4.6)
B (S)	1 or 2	9	1.35 (34.3) 1.18 (30.0)	1.312 (12.70)	.437 (11.09)	1.000 (25.40)	.10 (2.5)
C (T)	1 or 2	14	1.35 (34.3) 1.18 (30.0)	1.250 (31.75)	.500 (12.70)	.938 (23.82)	.19 (4.8)
D (U)	1 or 2	20	1.35 (34.3) 1.18 (30.0)	1.562 (39.67)	.500 (12.70)	1.250 (31.75)	.22 (55.9)
E (V)	1 or 2	26	1.35 (34.3) 1.18 (30.0)	1.625 (41.27)	.640 (16.25)	1.312 (33.32)	.22 (55.9)
F (W) or Q (X)	1 or 2	34	1.35 (34.3) 1.22 (31.0)	2.000 (50.80)	.828 (21.03)	1.688 (42.87)	.34 (8.6)
G (Y) or P (Z)	1 or 2	42	1.35 (34.3) 1.22 (31.0)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.34 (8.6)
H (1) or N (2)	1 or 2	50	1.35 (34.3) 1.22 (31.0)	2.312 (58.72)	.828 (21.03)	2.282 (57.96)	.34 (8.6
J (3)	1 or 2	75	1.35 (34.3) 1.22 (31.0)	2.594 (65.88)	1.188 (30.17)	2.282 (57.96)	.55 (14.0)
M (4)	1 or 2	75	1.35 (34.3) 1.22 (31.0)	2.594 (65.88)	1.188 (30.17)	2.282 (57.96)	.55 (14.0)

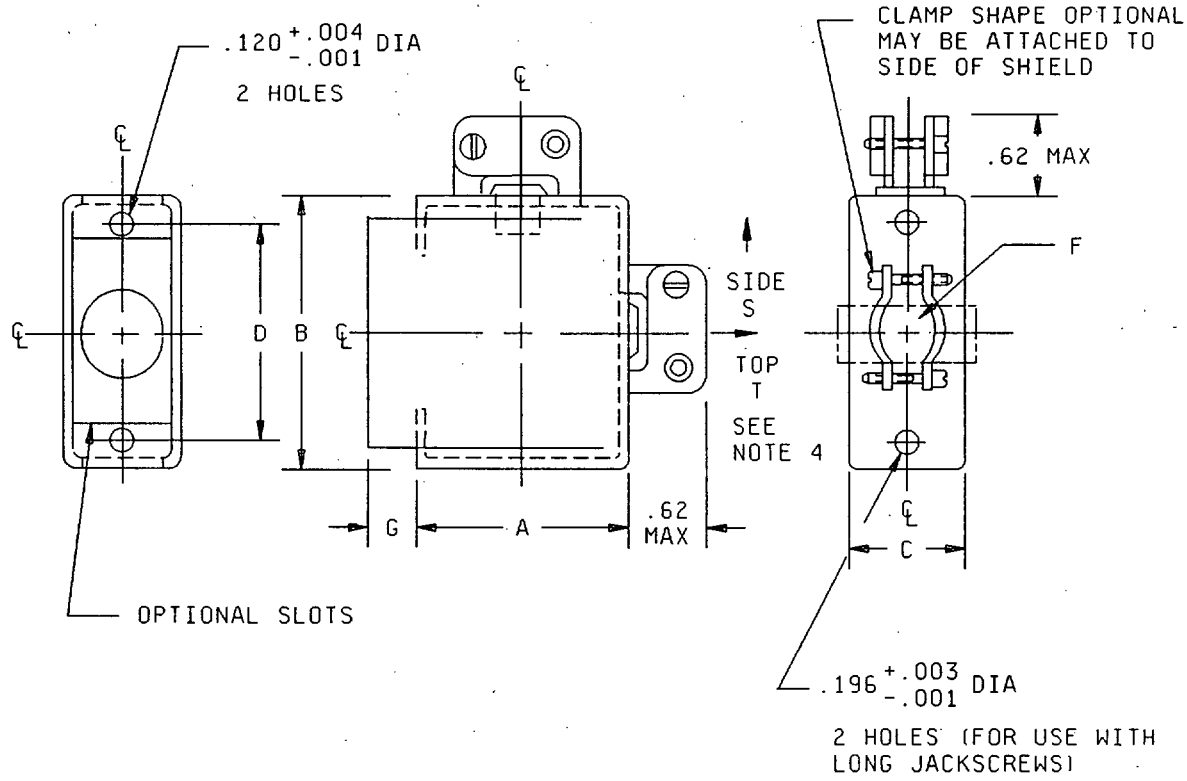
1/ Used when shell is specified.

2/ Insert designators L (6), M (4), N (2), P (Z), and Q (X) use with one piece jackscrews.

3/ Insert designators for piece part orders are in parentheses.

FIGURE 6. Shield configurations - Continued.

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Check the source to verify that this is the current version before use.



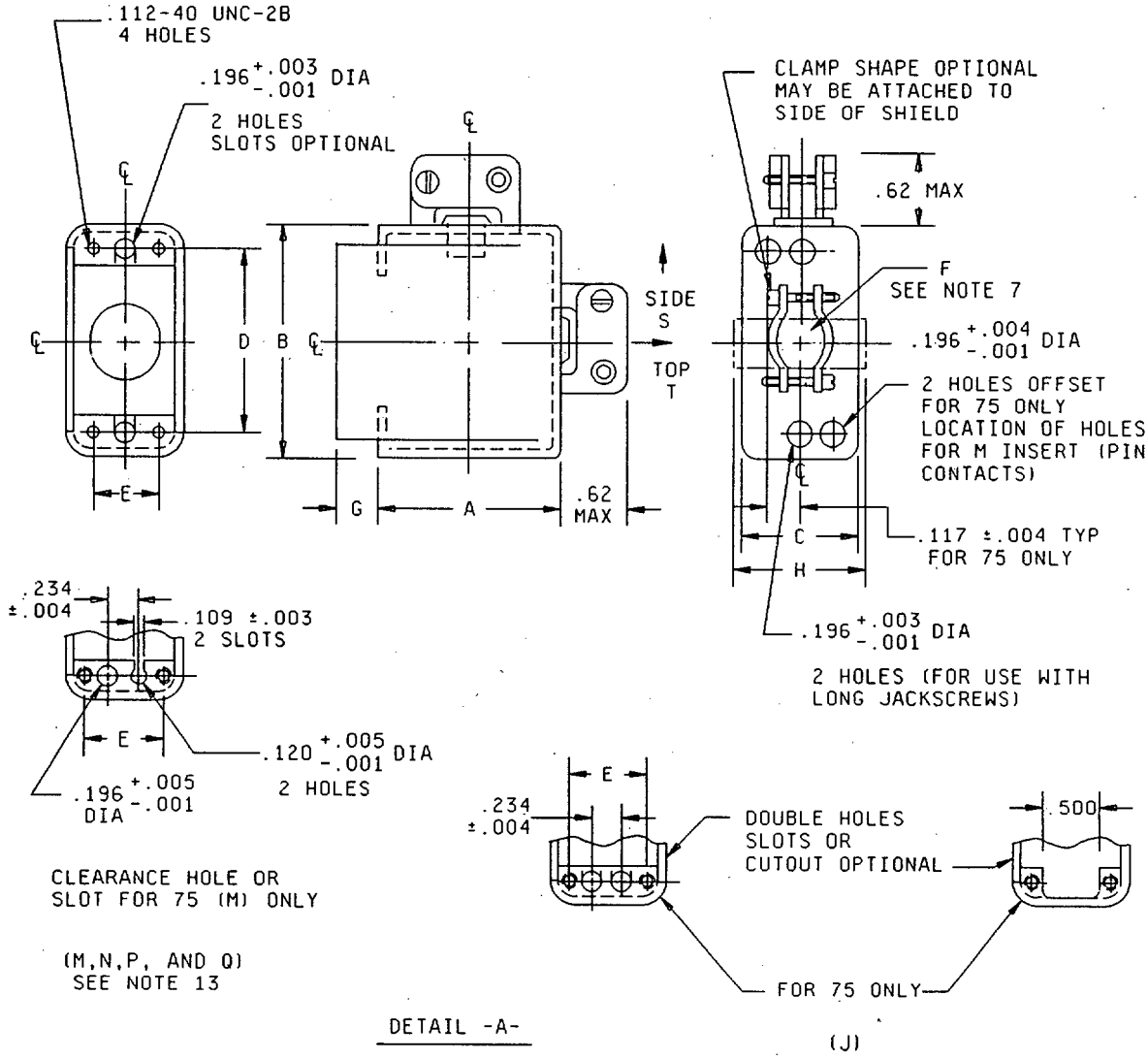
For use with insert configurations A (R), B (S), C (T), D (U), and E (V)

Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



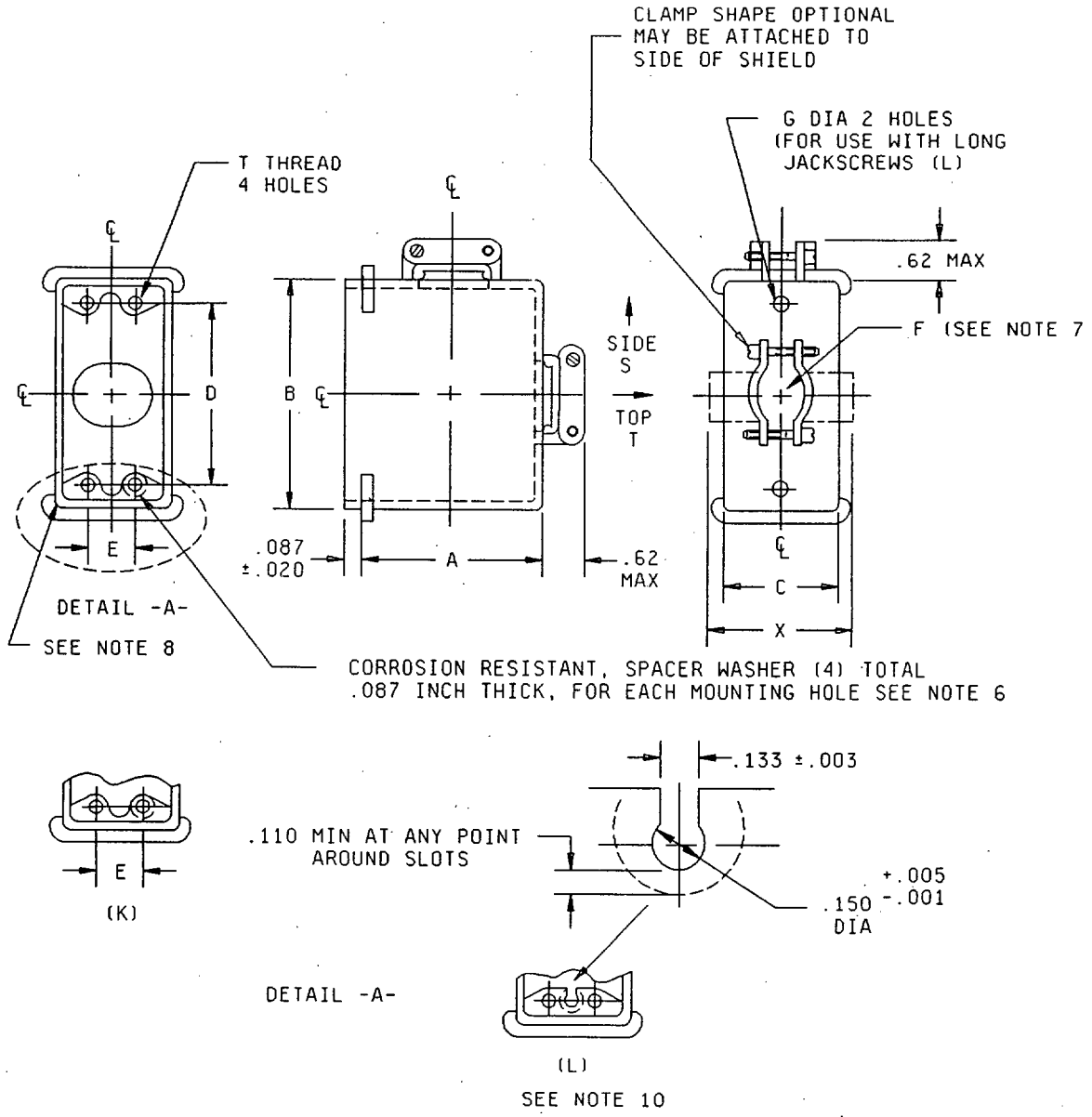
For use with insert configurations F (W), G (Y), H (1), J (3), M (4), N (2), P (Z), and Q (X)

Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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For use with insert configurations K (5) and L (6)

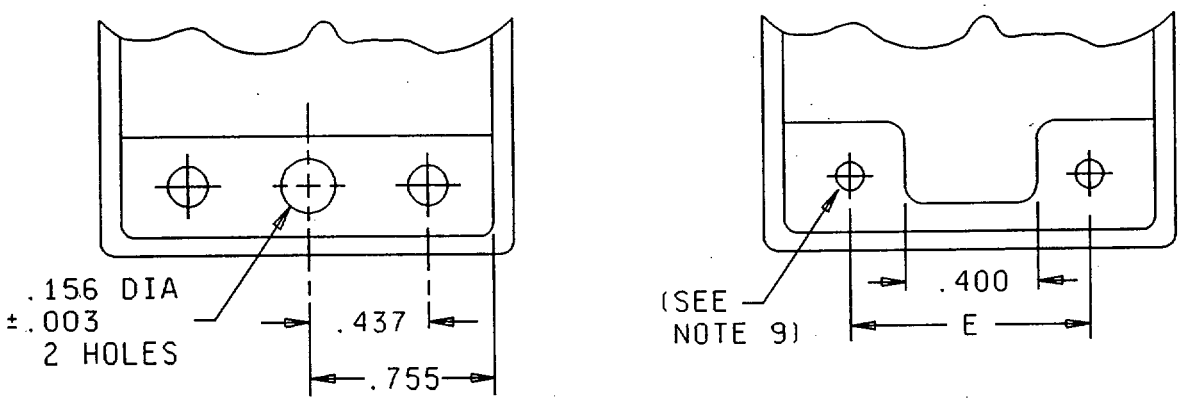
Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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DETAIL A



Alternate configurations K (5) and L (G)

For use with insert configurations K (5) and L (G)

Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Insert designator 1/ 2/	Configuration and cable opening	Connector size	A max min	B ±.016 (0.41)	C ±.016 (0.41)	D ±.005 (0.13)	F min cable opening	G ±.047 (1.19)
A (R) 3/	3 or 4	7	1.35 (34.3) 1.18 (30.0)	1.218 (32.51)	.437 (11.10)	.906 (23.01)	0.07 (1.8)	0.265 (6.73)
B (S) 3/	3 or 4	9	1.35 (34.3) 1.18 (30.0)	1.312 (33.32)	.437 (11.10)	1.000 (25.40)	0.16 (4.1)	0.265 (6.73)
C (T) 3/	3 or 4	14	1.35 (34.3) 1.18 (30.0)	1.250 (31.75)	.500 (12.70)	.937 (23.80)	.019 (4.82)	0.265 (6.73)
D (U) 3/	3 or 4	20	1.35 (34.3) 1.18 (30.0)	1.562 (39.67)	.500 (12.70)	1.250 (31.75)	.22 (5.6)	0.265 (6.73)
E (V) 3/	3 or 4	26	1.35 (34.3) 1.18 (30.0)	1.625 (41.28)	.634 (16.10)	1.312 (33.32)	0.26 (6.6)	0.265 (6.73)
F (W) 3/	3 or 4	34	1.35 (34.3) 1.22 (31.0)	2.000 (50.80)	.828 (21.03)	1.688 (42.88)	0.34 (8.6)	0.265 (6.73)
G (Y) 3/	3 or 4	42	1.35 (34.3) 1.22 (31.0)	2.312 (58.92)	.828 (21.03)	2.000 (50.80)	0.34 (8.6)	0.093 (2.362)
H (I) 3/	3 or 4	50	1.35 (34.3) 1.22 (31.0)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	0.34 (8.6)	.093 (2.36)
J (3) 3/	3 or 4	75	1.35 (34.3) 1.22 (31.0)	2.594 (65.89)	1.188 (30.18)	2.282 (57.96)	0.55 (14.0)	0.93 (2.4)
K (5) 3/ 4/ 5/	3 or 4	104	2.26 (57.4) 2.20 (55.9)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33) 6/	1.10 (2.8)	0.229 (5.81)
L (G) 3/ 4/ 6/	3 or 4	104	2.26 (57.4) 2.20 (55.9)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33) 6/	1.10 (2.8)	0.229 (5.81)

Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Insert designator 1/ 2/	Configuration and cable opening	Connector size	A max min	B ±.016 (0.41)	C ±.016 (0.41)	D ±.005 (0.13)	E ±.005 (0.13)	F min cable opening	G ±.047 (1.19)	H max
Q (X) 7/	3 or 4	34	1.35 (34.3) 1.22 (31.0)	2.000 (50.80)	.828 (21.03)	1.688 (42.87)	.468 (11.88)	.34 (8.6)	.265 (6.73)	1.13 (28.7)
P (Z) 7/	3 or 4	42	1.35 (34.3) 1.22 (31.0)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.88)	.34 (8.6)	.093 (2.36)	1.13 (28.7)
N (2) 7/	3 or 4	50	1.35 (34.3) 1.22 (31.0)	2.594 (65.88)	.828 (21.03)	2.282 (57.96)	.468 (11.88)	.34 (8.6)	.093 (2.36)	1.13 (28.7)
M (4) 7/	3 or 4	75	1.35 (34.3) 1.22 (31.0)	2.594 (65.88)	1.188 (30.17)	2.282 (57.96)	.766 (19.45)	.766 (19.45)	.093 (2.36)	1.37 (34.8)

1/ Insert designators for piece parts are in parentheses.

2/ Tolerance is ±.010 inch (0.25 mm), unless otherwise specified.

3/ For use when no shell is specified.

4/ For use with or without shells.

5/ T thread: .138-32 UNC-2B X max 1.88.

6/ T thread: .138-40 UNC-2B X max 1.88.

7/ Use with one-piece jackscrews and are not used with shells.

Shield

Configuration B

FIGURE 6. Shield configurations - Continued.

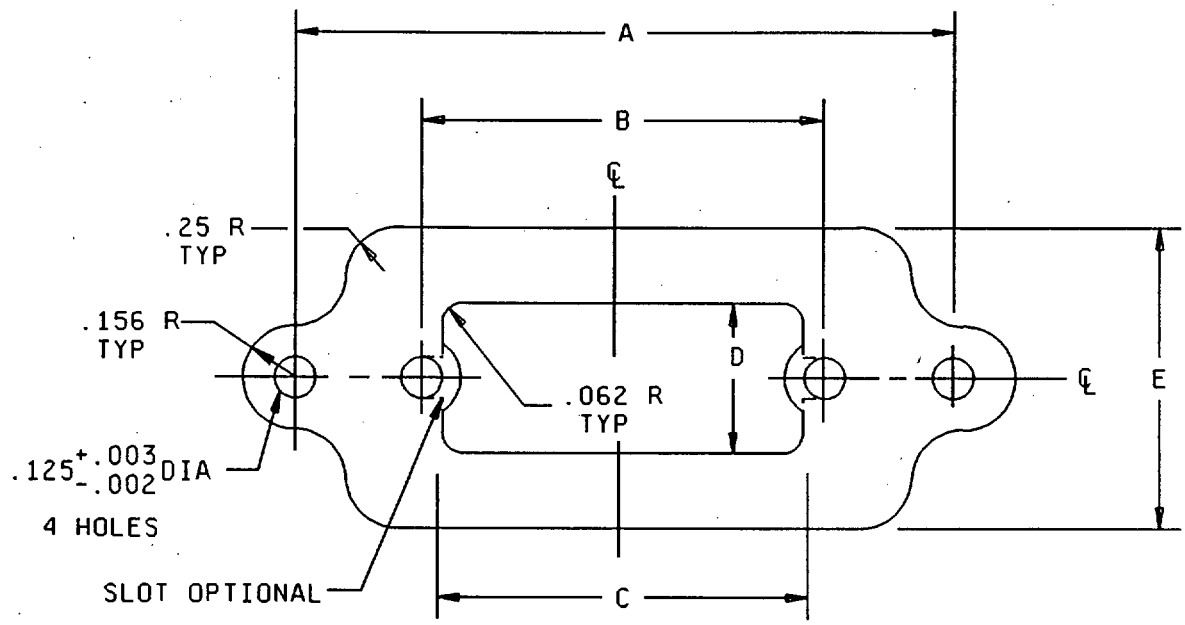
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.133	3.38	.550	13.97	1.250	31.75
.003	0.08	.138	3.51	.620	15.75	1.312	32.42
.004	0.10	.150	3.81	.755	19.17	1.350	34.29
.005	0.13	.160	4.06	.766	19.45	1.370	34.79
.010	0.25	.190	4.83	.828	21.03	1.562	39.67
.016	0.41	.196	4.98	.875	22 23	1.625	41.28
.020	0.51	.220	5.59	.890	22.61	1.660	42.16
.030	0.76	.226	6.60	.906	23.01	1.688	42.88
.047	1.19	.229	5.81	.937	23.80	1.969	50.01
.062	1.57	.234	5.94	1.000	25.40	2.000	50.80
.070	1.78	.260	6.60	1.100	27.94	2.230	56.66
.087	2.21	.265	6.73	1.110	28.19	2.282	57.97
.093	2.36	.340	8.63	1.130	28.70	2.312	58.72
.109	2.77	.400	10.16	1.170	29.72	2.366	60.10
.110	2.79	.437	11.09	1.180	29.97	2.375	60.33
.112	2.84	.460	11.70	1.211	30.76	2.594	65.89
.117	2.97	.468	11.89	1.218	30.94	2.835	72.01
.120	3.05	.500	12.70	1.220	30.98		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
3. Metric equivalents are in parentheses and are given for general information only.
4. Material: Aluminum alloy in accordance with QQ-A-250/8, temper H32, .040 inch (0.41 mm).
5. Finish: Anodized in accordance with MIL-A-8625, type I or type II.
6. Spacer washers (4), are used only with shell; discard in other uses. Aluminum shall be in accordance with QQ-A-250/11, T6, anodized in accordance with MIL-A-8625, type I or type II.
7. Edges of cable entry hole to be rounded to prevent insulation chafing.
8. No cracks, chips, or sharp edges on all 4 edges of flange.
9. Number 6-32 UNC-2B swedge and thread 4 holes.
10. For use with one-piece jackscrews.
11. Clamp hinge may attach to side of shield.
12. Contact type and jack screw orientation must be separated to guarantee intermateability.
13. Shield configuration B; insert designators M, N, P, and Q are for use with one-piece jackscrews and are not used with shell.

FIGURE 6. Shield configurations - Continued.

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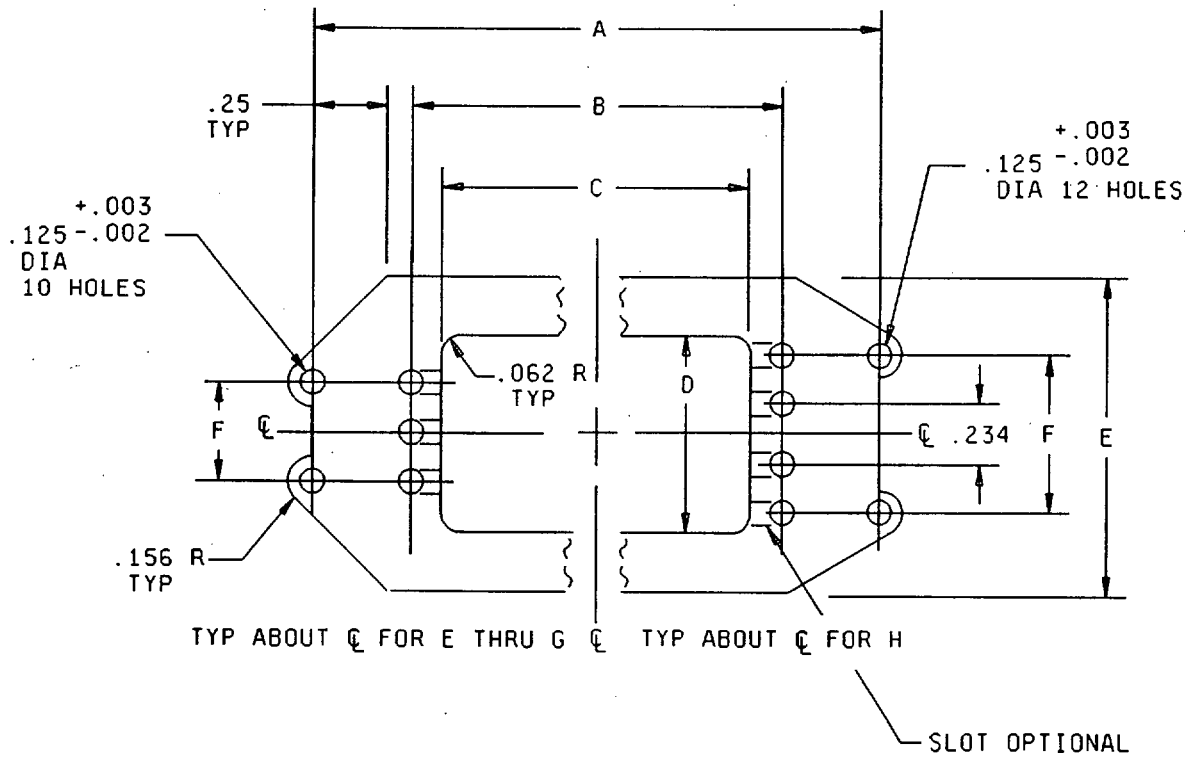
Retaining plate designator	Connector size (number of contacts)	A ±.005 (0.13)	B ±.005 (0.13)	C	D	E
A	9	2.016 (51.20)	1.000 (25.40)	.875 (22.22)	.406 (10.31)	.953 (24.20)
B	14	1.953 (49.60)	.937 (23.79)	.812 (20.62)	.469 (11.91)	1.016 (25.80)
C	20	2.266 (57.55)	1.250 (31.75)	1.125 (28.57)	.469 (11.91)	1.016 (25.80)
D	26	2.328 (59.13)	1.312 (33.32)	1.187 (30.14)	.593 (15.06)	1.141 (28.98)

Retaining plate

Configuration A

FIGURE 7. Shield retaining plate configurations.

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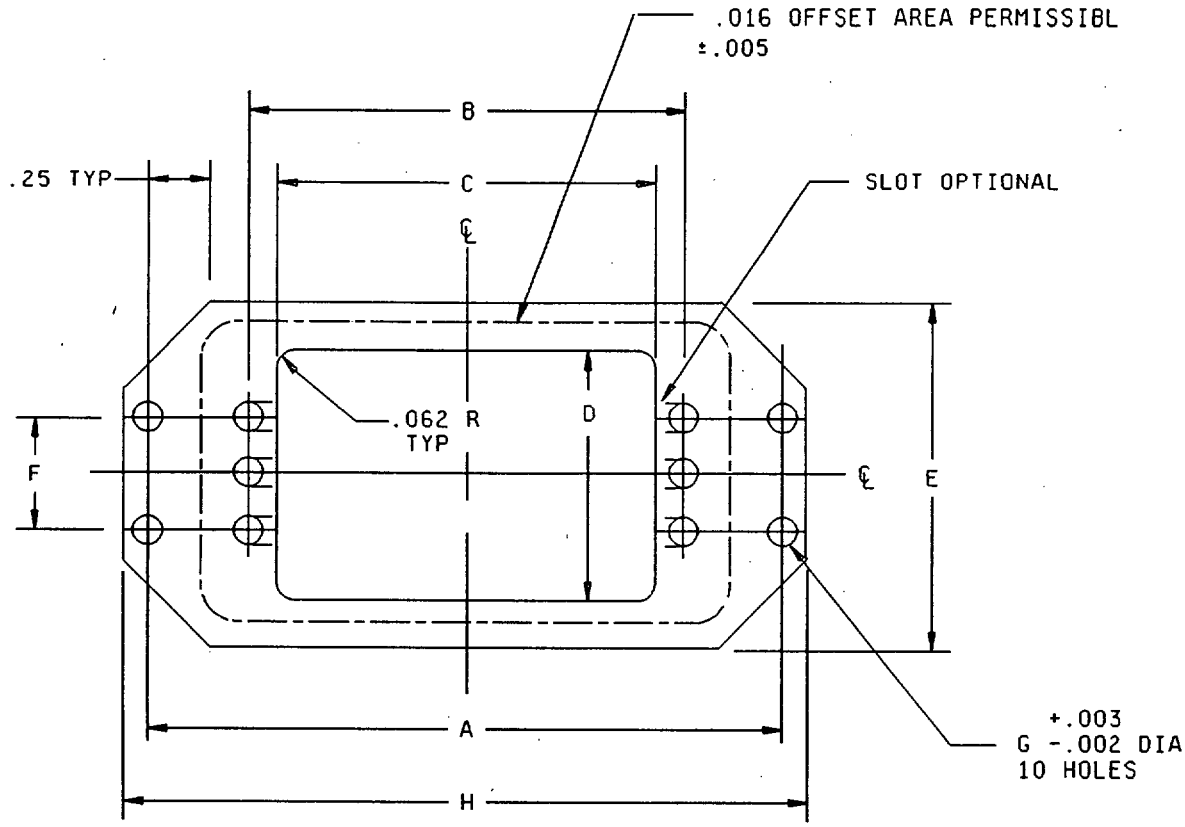
Retainer plate designator	Connector size (number of contacts)	A ±.005 (0.13)	B ±.005 (0.13)	C	D	E	F ±.005 (0.13)
F	34	2.703 (68.65)	1.687 (42.84)	1.438 (36.52)	.750 (19.05)	1.328 (33.73)	.468 (11.88)
G	42	3.016 (76.60)	2.000 (50.80)	1.750 (44.45)	.750 (19.05)	1.328 (33.73)	.468 (11.88)
H	50	3.422 (86.91)	2.281 (57.93)	2.031 (51.58)	.750 (19.05)	1.453 (36.90)	.468 (11.88)
J	75	3.422 (86.91)	2.281 (57.93)	2.031 (51.58)	1.109 (28.16)	1.797 (45.64)	.766 (19.45)

Retaining plate

Configuration B

FIGURE 7. Shield retaining plate configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Retainer plate designator	Connector size (number of contacts)	A ±.005 (0.13	B ±.005 (0.13)	C	D	E	F ±.005 (0.13)	G	H
K	104	3.500 (88.90)	2.375 (60.32)	2.125 (53.97)	1.486 (37.74)	2.000 (50.80)	.875 (22.25)	10.150 (3.81)	3.875 (98.42)

Retaining plate

Configuration C

FIGURE 7. Shield retaining plate configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.750	19.05	1.328	33.73	2.281	57.94
.003	0.08	.766	19.46	1.438	36.53	2.328	59.13
.005	0.13	.812	20.62	1.453	36.91	2.375	60.33
.016	0.41	.875	22.23	1.486	37.74	2.703	68.66
.062	1.57	.937	23.80	1.687	42.85	2.984	75.79
.125	1.57	.953	24.21	1.688	42.88	3.106	78.89
.150	3.81	1.000	25.40	1.718	43.64	3.296	83.72
.156	3.96	1.016	25.81	1.750	44.45	3.422	86.92
.234	5.94	1.109	28.17	1.953	49.61	3.500	83.90
.250	6.35	1.125	28.58	2.000	50.80	3.875	98.43
.406	10.31	1.141	28.98	2.016	51.21		
.468	11.89	1.187	30.15	2.031	51.60		
.500	17.70	1.250	31.75	2.125	53.98		
.593	15.06	1.312	33.32	2.266	57.56		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
4. Material: Aluminum alloy in accordance with QQ-A-250/8, temper H-32, .040 inch (1.02 mm) thick.
5. Finish: Anodize in accordance with MIL-A-8625, type I or type II.

FIGURE 7. Shield retaining plate configurations - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, 4, 5, 6, and 7.

Mating connector: See MIL-C-28748/4 and MIL-C-28748/6.

Nuts: As specified in guide pin configuration B. Not required with shield.

Part number: As shown in the following example: 1/

(See part number code)

M28748/5	A	C	1	H	L	1A
Basic number of specification sheet	Shell hardware 2/ designator (see figure 4 for configuration styles)	Insert designator (see figure 2)	Shield and shield clamp location or retaining plate (see figures 6 and 7)	Shell type and 3/ polarization (see figure 5)	Jackscrews or guidepins (see figure 1)	Contacts 4/

PART NUMBER CODE:

Shell hardware 2/ designators	Insert designators	Number of contacts positions
A- long jackscrew hardware only	A	7
B- all other hardware	B	9
-- (insert only)	C	14
	D	20
	E	26
	F	34
	G	42
	H	50
	J	75
	K	104
	O - No insert included	

Insert 7/ Designators through hole	
Q	34
P	42
N	50
M	75
L	104

Insert 8/ designators for piece part orders	
R	7
S	9
T	14
U	20
V	26
W	34
Y	42
1	50
3	75
5	104

Shield and 5/6/ shield clamp location or retaining plate	
1--top opening	
2--side opening	
3--top opening	
4--side opening	
A--	
B |	
C |	
D |-Retaining	
E | plate	
F |	
G |	
H |	
J |	
K--	
O--none included	

Shell type and polarization designators		Polarization positions 3/
A--		A
B |		B
C |		C
D |-Receptacle		D
E | shells		E
F |		F
G--		G
H---Unpolarized		H
J--		A
K |		B
L |		C
M |-Plug		D
N | shells		E
P |		F
Q--		G
R--Unpolarized		H
O--No shell included		

Through hole 7/ 8/ Insert designators for piece part orders	
|	
|	
|	
|	
|	
X	
Z	
2	
4	
6	

Contacts: Per MIL-C-28748
1A - 100 percent size 20
1L - None included

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Jackscrews
or
guidepins
M - long jackscrews, multipiece construction, configuration E, figure 1.
L - long jackscrews, one piece construction, configuration F, figure 1.
S - short jackscrews, configuration D, figure 1 (not used with shield).
F - fixed jackscrews, configuration C, figure 1 (not used with shield).
G - guidepins, configuration B Figure 1.
0 - none included.1/

Jack and guide location: The male jackscrew (turning or fixed) or male guidepin shall be located by the "A" or "1" socket contact.

1/ The number zero (0) is used to indicate which parts are not included.

2/ Not applicable to size 7 insert designator A.

3/ Polarization positions are for location of plug and receptacle shell pins and slots (see figure 4).

4/ Contacts: In accordance with MIL-C-28748, 1A - 100 percent size 20. To order without contacts use the designator, 1L.

5/ Screws: MS35275-215 furnished with shield configurations; insert designators when required. MS35275-229 furnished with shield configurations; insert designators when required. Cross drilled hole is optional.

6/ Shield configuration A is 1 and 2, and shield configuration B is 3 and 4.

7/ Through hole insert configurations are used with long jackscrews only.

8/ These designators are informational and used for insert reference (connector size) to order an appropriate shield or shell as a piece part.

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TABLE I. Superseded specification and MS sheets.

Specification or MS sheet	Date
MIL-C-28748/5	30 November 1981
MS18192	1 June 1983
MS18193	14 November 1988
MS18194	11 April 1974
MS18195	20 may 1970
MS18196	23 June 1972
MS18197	23 June 1972
MS18198	23 June 1972
MS18199	23 June 1972
MS18200	20 May 1970
MS18203	26 April 1990
MS18204	20 December 1965
MS18257	23 June 1972
MS18258	23 June 1972
MS18261	23 June 1972
MS18263	23 June 1972
MS24004	23 June 1972
MS24010	23 June 1972
MS24014	23 June 1972
MS24020	23 June 1972
MS24022	23 June 1972
MS24026	23 June 1972
MS24132	1 June 1976
MS24133	3 January 1979

Revision letters are not used due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV, ME, AR
Navy - MC, OS, SH

Preparing activity
Navy - EC

Agent:
DLA - ES

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